Computers and Education: Artificial Intelligence 10 (2026) 100570

Contents lists available at ScienceDirect

Computers and Education: Artificial Intelligence

journal homepage: www.sciencedirect.com/journal/computers-and-education-artificial-intelligence

A systematic review of the impact of GenAI on learning performance, AI hallucinations, and problem-solving in computer science education

Adedeji Adefisoye Adejumo [a,b], Solomon Sunday Oyelere [c,d,*], Ismaila Temitayo Sanusi [a], Jarkko Suhonen [a]

[a] University of Eastern Finland, School of Computing, P.O. Box 111, FI-80101, Joensuu, Finland
[b] Modibbo Adama University Yola, PMB 2076 Yola, Adamawa State, Nigeria
[c] Department of Computer Science, University of Exeter, Exeter, EX4 4SB, United Kingdom
[d] Department of Computer Science, Electrical and Space Engineering, Luleå University of Technology, SE-971 87, Luleå, Sweden

A R T I C L E I N F O

Keywords:
Generative AI
Computer Science Education
AI Hallucinations
Adaptive Learning
Metacognition
Problem-Solving Skills
Learning Performance

A B S T R A C T

This systematic review synthesizes 64 empirical studies to examine how Generative AI (GenAI) shapes learning in Computer Science Education (CSE), particularly in programming, debugging, algorithmic reasoning, and computational problem-solving contexts. Grounded in Constructivist, Sociocultural, Cognitive Load, Adaptive Learning, and Metacognitive Learning theories, the review adopts an integrative perspective to analyze how GenAI-driven adaptivity, AI output qualities, hallucination dynamics, and cognitive–affective regulation influence learners' interpretation, cognitive processing, and learning outcomes. Findings reveal a dual impact of GenAI in CSE. On the negative side, hallucinated or misleading outputs can increase extraneous cognitive load during programming and debugging and promote over-reliance on system-generated content. They may also perpetuate inequities due to limited access in low-resource settings or insufficient support for culturally and linguistically diverse learners. These effects can disrupt error detection, self-monitoring, and problem-solving, leading to impaired learning performance and widened educational disparities. On the positive side, when embedded within structured, equitable, and pedagogically grounded environments, GenAI supports reflective programming practice by promoting self-monitoring, verification, and strategic adjustment, thereby enhancing problem-solving skills, engagement, and personalized learning outcomes. By framing learning performance, hallucination dynamics, and problem-solving as interconnected dimensions of GenAI-supported computing education, this review provides a theoretically coherent and pedagogically grounded lens for understanding how GenAI reshapes learning in CSE. The review's novelty lies in its integrative conceptual framework, offering actionable insights for designing equitable, cognitively balanced, and instructionally effective GenAI-supported learning environments.

1. Introduction

Generative AI (GenAI) has emerged as a significant trend in Artificial Intelligence, rapidly gaining global attention. Its remarkable capabilities have raised important questions within the computing community about how instructors should adopt and integrate these technologies into their pedagogy (Prather, Denny, et al., 2023). Educators are increasingly using GenAI to support adaptive and personalized learning, transforming educational practices in computer science education (Guettala et al., 2024; Prather, Leinonen, et al., 2025). However, despite these promising prospects, AI hallucinations undermine the credibility and reliability of GenAI-supported learning systems (Kim & Lee, 2024). The generation of deceptive or false information can negatively affect learning outcomes and cognitive processes (Zhai et al., 2024). These outputs disrupt students' ability to monitor understanding, detect errors, and adjust strategies essential for effective learning and problem-solving.

Recent curriculum authorities emphasize that GenAI represents a transformative moment for Computer Science Education (CSE). The ACM/IEEE CS2023 task force notes that emerging AI technologies are expected to reshape course content, pedagogy, and assessment practices, while also posing major integration challenges (Kumar et al., 2024). This

* Corresponding author. Luleå University of Technology, SE-971 87, Luleå, Sweden.
E-mail address: solomon.oyelere@ltu.se (S.S. Oyelere).

https://doi.org/10.1016/j.caeai.2026.100570
Received 17 July 2025; Received in revised form 28 February 2026; Accepted 5 March 2026
Available online 14 March 2026

underscores the urgent need to examine not only the opportunities but also the risks GenAI introduces in computing learning environments, particularly regarding feedback accuracy, equitable personalization, and learner regulation (National Academies of Sciences et al., 2021). Despite offering adaptive instructional support, GenAI presents risks such as hallucinations and inconsistent guidance that can undermine learning. These concerns highlight the need for careful and ethically informed integration, as emphasized by an ACM task force on the societal impacts of GenAI in higher computing education (Mak et al., 2025).

According to the ACM/IEEE CS2023 Curricula guidelines, CSE encompasses the theories, pedagogies, and technologies involved in teaching computing concepts, programming, problem-solving, and computational thinking. It includes both formal and informal learning environments and emphasizes computing principles, algorithmic reasoning, software development, and ethical practices (Kumar et al., 2024). Within this definition, CSE extends beyond disciplinary content. It includes the learning processes through which students engage with problem-solving, feedback, and adaptive instructional technologies. This perspective supports examining how GenAI-mediated personalization, hallucination dynamics, cultural context, and adaptive support shape learning performance and problem-solving outcomes in CSE.

Although GenAI tools have applications across various disciplines, their role in CSE is uniquely shaped by the field's cognitive and procedural demands (Sands, 2019). Learning programming requires mastering syntax, algorithmic reasoning, abstraction, and iterative problem-solving and debugging. All these tasks demand sustained cognitive effort and regulation (Susanti et al., 2021). Consequently, GenAI's impact in CSE extends beyond generic educational support, directly influencing students' ability to design algorithms, write correct code, and solve computational problems. A recent systematic review by Agbo et al. (2025) shows that GenAI tools are being explored across computing education with varying effects on learning outcomes, pedagogy, and student engagement. This diversity suggests that effective GenAI integration must carefully account for the unique cognitive and procedural demands of computing education.

The educational success of students depends on personal, social, cultural, and institutional factors (Dechavez, 2024). Although GenAI technologies contribute to advancing personalized and adaptive learning, a notable gap remains in understanding how these platforms address cultural diversity and inclusivity, particularly in CSE. Culturally responsive teaching is essential for acknowledging students' varied backgrounds and creating inclusive learning environments that support equity and social justice (Samuels, 2018). However, GenAI introduces risks, including erroneous or hallucinated outputs, which can mislead learners, promote ineffective solutions, and negatively affect learning outcomes (Ahmad et al., 2023; Leinonen et al., 2023).

Several studies have explored GenAI applications in CSE, particularly for enhancing personalized learning, student engagement, and programming support. For example, Codex has been used to generate code and explain errors, supporting novice programmers' conceptual understanding and problem-solving (Finnie-Ansley et al., 2022; Leinonen et al., 2023). Similarly, GenAI-powered Intelligent Tutoring Systems provide dynamic content, immediate feedback, and tailored learning pathways, improving engagement and learning outcomes (Liu, Guo, et al., 2024; Maity & Deroy, 2024). Other work has examined GenAI for feedback on open-ended responses (Matelsky et al., 2023), programming assistance (Pankiewicz & Baker, 2023; Sarsa et al., 2022), and broader systematic reviews of its educational role (Ogunleye et al., 2024; Yusuf et al., 2024). However, few reviews have systematically investigated how GenAI influences learning performance, hallucinations, and problem-solving in CSE, particularly in the context of cultural diversity and critical thinking. This gap highlights the need for an integrative review to synthesize empirical evidence and clarify GenAI's impact on learning outcomes in CSE.

To address this gap, this study adopts an integrative perspective to examine how GenAI tools influence learning processes in CSE. Rather than treating personalized learning, hallucination dynamics, problem-solving, and learning outcomes as isolated constructs, this review conceptualizes them as interacting mechanisms within a unified framework. Within this framework, GenAI-driven adaptivity and hallucination-induced uncertainty are associated with variations in metacognitive regulation. These processes include self-monitoring and confidence judgment (Liu et al., 2025), as learners evaluate their understanding and regulate reliance on GenAI outputs (Lan & Zhou, 2025; Tankelevitch et al., 2024); error detection when identifying inconsistencies or inaccuracies in AI-generated responses and code corrections (Hoq et al., 2025; Kohen-Vacs et al., 2025); and cognitive disruption (Ma et al., 2025). These mechanisms relate to differences in learning performance, problem-solving skills, and critical thinking outcomes. By framing these dimensions as interdependent components of GenAI-supported learning, the study provides a theoretically coherent lens for understanding how AI-mediated personalization and misinformation jointly affects student cognition, engagement, and achievement. Accordingly, this systematic review does not seek to establish causal relationships between GenAI use and learning outcomes. Instead, it synthesizes empirical evidence to identify patterns of association, interaction, and contextual influence across personalized learning, hallucination dynamics, cultural inclusivity, and problem-solving in CSE.

Despite the growing adoption of GenAI in CSE, significant pedagogical challenges remain, including high cognitive load during programming and debugging (Huang et al., 2025), automation bias and over-reliance on AI feedback (Feng et al., 2025), hallucinated outputs (Reihanian et al., 2024), uneven prior knowledge among learners, including differences in factual knowledge (Qi et al., 2025), and cultural and linguistic variability (Prather, Reeves, et al., 2025). These challenges can undermine problem-solving, learning performance, and equitable engagement. To interpret and address these issues, this review draws on established learning theories, including Constructivist, Sociocultural, Cognitive Load, Adaptive Learning, and Metacognitive Learning theories, as explanatory lenses. In the subsequent sections, we demonstrate how each theory clarifies learners' interactions with GenAI, guides adaptive feedback, and illuminates mechanisms by which AI can both support and hinder effective learning.

This review also examines how GenAI tools support culturally diverse learners through adaptive and inclusive strategies. Such strategies aim to bridge linguistic and contextual gaps and shape how learners interpret, verify, and act upon AI-generated information. By focusing on these aspects, the study clarifies how GenAI-mediated learning influences performance, hallucination handling, and problem-solving in CSE. To ensure rigor, the review follows PRISMA and Kitchenham guidelines (Kitchenham & Charters, 2007; Page et al., 2021) for systematic reviews, covering four major databases (Scopus, Web of Science, IEEE Xplore, and ACM Digital Library) for comprehensive and relevant CSE literature (Falagas et al., 2008; Luxton-Reilly et al., 2018; Martín-Martín et al., 2021).

To achieve the research objectives, we formulated the following key research questions prior to conducting the systematic search:

RQ1: What are the impacts of GenAI-based personalized learning tools on academic performance in computer science education?
RQ2: How do GenAI hallucinations influence cognitive processes and learning outcomes in computer science education?
RQ3: How do GenAI platforms support culturally diverse learners' cognitive engagement in computer science education?
RQ4: How do adaptive learning tools driven by GenAI influence problem-solving and critical thinking skills in computer science students?

These research questions provide a lens for synthesizing evidence on learning performance, hallucination dynamics, cultural context, and problem-solving in GenAI-enhanced CSE, guiding the review of empirical studies across these dimensions. The remainder of the paper is organized as follows: Section 2 reviews literature on GenAI-supported personalized learning, cultural diversity, and hallucination dynamics in CSE; Section 3 presents the systematic review methodology; Section 4 reports the findings, which are discussed in Section 5; and Section 6 concludes with implications, limitations, and directions for future research.

2. Background

GenAI has brought about a growing body of research in CSE. Researchers are examining its impact on personalized and adaptive learning, cognitive processes, and issues related to bias, cultural inclusivity, and hallucinations. Tools such as ChatGPT, Gemini, Claude, and Copilot have been widely studied, particularly for their roles in programming education and their potential to improve learning outcomes and learner experiences. At the same time, concerns persist regarding biased or misleading AI-generated content. This section reviews existing literature on GenAI in CSE, focusing on personalized and adaptive learning, cultural diversity and inclusivity, and AI hallucinations. Across these themes, recent studies suggest that GenAI's educational impact is mediated by learners' metacognitive regulation, specifically how students monitor, interpret, verify, and act upon AI-generated outputs during learning and problem-solving.

2.1. Impact of GenAI-based personalized and adaptive learning on academic performance

GenAI-based personalized and adaptive learning influences academic performance in CSE primarily by shaping learners' cognitive processes during problem-solving, including self-monitoring, strategy selection, confidence calibration, and cognitive load management. The integration of GenAI into CSE has transformed personalized learning (Zimmermann et al., 2024), enhancing learning experiences, motivation, and interest among educators and students alike (Deriba et al., 2024). Personalized learning is a learner-centered approach that considers students' needs, preferences, and capabilities. It aims to adapt instructional pacing, feedback, and task difficulty in ways that support learners' reflection and strategy adjustment (Fariani et al., 2023). Adaptive learning is a form of personalized learning that leverages computer-based systems and AI to provide data-driven instruction, dynamically adjusting content based on learners' prior knowledge and performance (Taylor et al., 2021, pp. 17–34).

The impact of GenAI-based personalized and adaptive learning on academic performance in CSE can be interpreted through established learning theories. The theories explain how GenAI addresses core challenges such as high cognitive load, uneven prior knowledge, and difficulties in debugging and strategy selection.

Adaptive Learning approaches (Tarun et al., 2025) explain how GenAI-driven systems dynamically adjust task difficulty, feedback, and pacing in response to learner performance. This helps to manage heterogeneity in prior knowledge while maintaining optimal challenge levels during programming and algorithmic reasoning tasks. Constructivist Learning Theory (Guo et al., 2024; Wu, 2023) accounts for how learners actively construct understanding through interaction with personalized explanations, examples, and problem-solving prompts, while also highlighting the risk that excessive automation may weaken conceptual engagement.

Cognitive Load Theory (Lo et al., 2024; Sweller, 2020) provides a basis for analyzing how GenAI-generated feedback and scaffolding influence intrinsic and extraneous load during programming and debugging. Well calibrated personalization can reduce unnecessary cognitive burden, whereas poorly structured or excessive assistance may increase load or promote surface-level learning. These theories clarify that improvements in academic performance arise not from personalization alone, but from how GenAI-supported adaptivity promote effective learning and problem-solving processes.

GenAI is transforming personalized and adaptive learning in CSE. Tools such as ChatGPT, GitHub Copilot, Codex, Gemini, Claude, and Mistral are increasingly embedded in learning systems to generate code examples, exercises, explanations, and assessments tailored to learners' skill levels, interests, and goals (Kurdi et al., 2020; Ouh et al., 2023; Rane et al., 2024; Sarsa et al., 2022). GenAI has also been applied to the automated generation of multiple-choice questions across varying proficiency levels and subject areas in computer science (Shoaib et al., 2025).

Beyond content personalization, these systems provide on-demand guidance, feedback, and instructional scaffolding, thereby reshaping how learners engage with problem-solving tasks. Learners' ability to evaluate and apply AI-provided guidance becomes central in determining whether personalization supports deep learning or merely surface-level performance gains. By analyzing student data and adapting instructional pathways to individual strengths and weaknesses, GenAI-powered systems enable self-paced learning while shaping learners' engagement and academic performance (Kamalov et al., 2023; Rane et al., 2024).

Research has shown that adaptive intelligent tutoring systems and automated code feedback tools developed using GenAI can accommodate individual learners and significantly improve understanding and learning outcomes in programming courses (Kwak et al., 2023; Zhang et al., 2024). For instance, Yan et al. (2024) designed a programming lesson that positioned GenAI as a learning companion to support collaboration between students and AI. The findings indicated that effective student–GenAI partnerships enhanced learners' cognitive engagement and self-directed learning, supporting autonomous problem-solving and sustained performance.

By integrating ChatGPT into existing eLearning platforms, Troussas et al. (2024) enabled instructors to generate individualized module assessments tailored to each student's profile. The evaluation showed that most instructors found the system user-friendly and reported significant time savings in assessment creation, underscoring the potential of GenAI-driven automation to support personalized programming education.

With GenAI's rapid integration into academic context, understanding its effect on cognitive skills is essential for developing effective teaching methods and improving learning outcomes (Zhou et al., 2024). Focusing on coding abilities, Mahaini (2024) examined the impact of GenAI on computer science students' learning outcomes, including engagement, motivation, and critical problem-solving skills. They observed performance gains associated with GenAI use, particularly when learners actively engaged in reflection and strategic problem-solving rather than passively accepting generated solutions.

Using ChatGPT to investigate the impact of GenAI adaptive learning on programming education, Yilmaz and Yilmaz (2023) examined 45 undergraduate students enrolled in a programming course. The findings revealed that the experimental group showed significantly higher computational thinking skills, self-efficacy, and motivation compared to the control group. However, recent systematic evidence also cautions that despite these benefits, GenAI tools may diminish foundational programming logic and higher-order thinking skills when used without structured pedagogy (Nathaniel et al., 2025b).

Leveraging Generative Pre-Trained Transformer (GPT) models, Crandall et al. (2023) developed an Automatic Review Tool (ART) to provide timely, AI-generated code reviews for computer science and software engineering students. The tool reduced tutors' workload while delivering prompt feedback to students. Integrated into a GitHub-based assignment submission platform, ART successfully detected defects and offered personalized suggestions. To prevent over-reliance on automated assistance, Liffiton et al. (2023) developed CodeHelp, a large language model–powered tool that guides students without revealing complete solutions. It provides real-time support while enabling instructors to monitor usage and learning progress in programming. These designs acknowledge that unchecked automation can undermine learning processes, particularly learners' ability to detect errors, sustain productive struggle, and develop robust problem-solving skills.

This synthesis extends prior reviews by reframing academic performance in GenAI-supported CSE as an outcome shaped by how learners engage with AI-supported personalization and feedback. Rather than treating personalization as inherently beneficial, the reviewed evidence indicates that performance gains and problem-solving development depend on whether GenAI tools support meaningful learning processes or encourage overreliance and cognitive offloading.

2.2. Cultural diversity and inclusivity in AI-powered computer science education

Access to quality CSE is essential for equitable societal development, yet disparities persist, particularly among marginalized and underserved populations (Goode et al., 2020). AI offers innovative opportunities to reduce these gaps by supporting personalized pedagogical strategies that promote equitable access to computing education. A key challenge for educators in multicultural classrooms is designing instruction and assessment that meaningfully reflect students' cultural and linguistic backgrounds (Kouo, 2022). Cultural and linguistic contexts shape how learners interpret, evaluate, and apply instructional feedback and problem-solving guidance, processes that become especially pronounced when such guidance is generated by GenAI systems.

GenAI tools are capable of producing and adapting content dynamically to tailor learning materials to individual student needs (Sandhu et al., 2024, pp. 1–28). For instance, GenAI-based adaptive learning platforms can incorporate culturally relevant examples and language, thereby reducing linguistic barriers and increasing the relatability of technical concepts (Anis, 2023).

The importance of cultural diversity and inclusivity in AI-powered CSE can be explained through Sociocultural Learning Theory (Coppens & Kelley, 2025), which emphasizes that learning is mediated by language, cultural tools, and social context. In GenAI-supported environments, this perspective helps explain why learners may struggle when AI-generated explanations or feedback are linguistically or culturally misaligned, and why culturally contextualized output can improve engagement, interpretability, and equity in learning outcomes.

Constructivist Learning Theory (Guo et al., 2024) further explains how learners integrate culturally meaningful examples into existing knowledge structures during programming and problem-solving, while highlighting the risk that decontextualized or unfamiliar AI-generated content may hinder conceptual understanding. Adaptive Learning approaches (Tarun et al., 2025) provide a rationale for how GenAI systems can dynamically adjust learning pathways, representations, and feedback to accommodate diverse learner profiles and uneven prior knowledge. These theories clarify that culturally responsive GenAI systems shape learners' cognitive engagement, reasoning, and judgment during problem-solving in CSE.

Embedding ethical principles and cultural inclusivity in the design, development, and deployment of GenAI systems in CSE is especially important, given the field's historical grounding in Western-centric paradigms (El-Sayed, 2023; Schelhowe, 2004). Despite growing awareness, implementing culturally responsive AI remains challenging due to persistent algorithmic bias. Large language models often reflect stereotypes, misrepresentations, and societal biases embedded in training data, model architectures, and user interactions, raising concerns about equity, diversity, and fairness (Yogarajan et al., 2025). Addressing these challenges requires incorporating diverse datasets, engaging cultural experts, and adopting inclusive design practices to ensure AI-generated content resonates with diverse learner populations.

Recent studies have developed culturally blended personalized GenAI solutions. For example, Nyaaba et al. (2024) introduced the GenAI Culturally Responsive Science Assessment (GenAI-CRSciA) framework to automate the creation of culturally contextualized assessment items for K–12 education, incorporating elements such as Indigenous knowledge, ethnicity, language, and religion. Similarly, Nyaaba and Zhai (2025) developed culturally responsive lesson planner (CRLP) custom GPTs using Interactive Semi-Automated (ISA) prompts to enhance digital diversity and inclusivity in global education. These tools generated culturally relevant lesson plans that integrated language and Indigenous knowledge more effectively than generic GPT outputs. However, both studies also reported instances of misconceptions and hallucinated local terms, requiring learners and educators to engage in verification, contextual reasoning, and critical evaluation to identify and correct inaccuracies.

This synthesis extends prior work by framing cultural diversity not merely as a design consideration, but as a contextual moderator that shapes learning performance, hallucination handling, and problem-solving in GenAI-supported CSE. The reviewed evidence indicates that culturally responsive GenAI tools influence how learners interpret AI-generated explanations, evaluate their reliability, manage hallucinated content, and apply feedback during problem-solving. As such, cultural inclusivity functions as a critical contextual factor shaping cognitive engagement, learning outcomes, and equity in GenAI-enhanced CSE.

2.3. Generative AI hallucinations in computer science education

Although LLMs exhibit remarkable capabilities across various applications, concerns remain about their tendency to hallucinate, particularly in educational contexts where learners must evaluate trust, accuracy, and problem-solving strategies when engaging with AI-generated outputs. Such models may generate content that deviates from user input, contradicts prior context, or misaligns with factual knowledge (Zhang et al., 2023).

Hallucination refers to the production of erroneous, fictional, or fabricated information by GenAI models, posing a substantial challenge when users rely on these systems for learning and critical decision-making (Feldman et al., 2023; Patel, 2024). These hallucinations often arise when models favor novelty over accuracy and relevance (Mukherjee & Chang, 2023), manifesting as incorrect code suggestions (Liu, Liu, et al., 2024), misleading explanations, or inconsistent feedback (Liang et al., 2024). Consequently, systems such as ChatGPT may present a blend of factual and non-factual content in a convincing manner, making it difficult for learners to accurately evaluate the credibility of generated outputs (Oviedo-Trespalacios et al., 2023).

From a learning-theoretical perspective, this challenge is particularly salient in CSE because hallucinated content interferes with how learners construct, evaluate, and apply knowledge during problem-solving. Constructivist learning theory posits that learners integrate new information with existing cognitive models (Harris et al., 2013). Hallucinated outputs disrupt this process by introducing erroneous elements into learners' conceptual frameworks, which can lead to misconceptions, unstable understanding, or flawed problem-solving strategies (Elsayed, 2024).

From the perspective of Cognitive Load Theory, hallucinated or misleading AI outputs increase extraneous cognitive load by forcing learners to reconcile conflicting or incorrect information, thereby diverting limited cognitive resources away from effective reasoning, debugging, and solution development (Elsayed, 2024; Yatani et al., 2024). As a result, learners may experience reduced focus, impaired comprehension, and diminished problem-solving efficiency. A prominent manifestation of hallucination in CSE is code hallucination, which occurs when code generated by LLMs appears syntactically valid but fails to execute correctly or meet task requirements. Such hallucinated code can lead to runtime errors, logical flaws, and incorrect program behavior, complicating the reliable integration of LLMs into programming education and software development (Tian et al., 2024).

Although AI hallucinations pose significant challenges in programming context, they may also trigger productive cognitive disruption by prompting learners to monitor understanding, detect errors, and adjust strategies (Asbai, 2024). From a theoretical perspective, these challenges may also function as catalysts for metacognitive engagement. According to Metacognition Theory (Tankelevitch et al., 2024; Xu et al., 2025), learners regulate their cognition through processes such as planning, monitoring, and evaluating. Hallucinations can therefore prompt students to reflect, self-monitor, and refine their strategies when interpreting or verifying GenAI outputs (Cai & Gao, 2025), positioning hallucination handling as a cognitive regulation process rather than merely an error condition.

Hallucinations are commonly categorized as intrinsic or extrinsic. Intrinsic hallucinations occur when generated content contradicts source information, whereas extrinsic hallucinations produce information unrelated to the source and neither supported nor refuted by it (Ji et al., 2023; Patel, 2024). These hallucinations may arise from factors such as overfitting, model complexity, and biases embedded in training data (Choudhury et al., 2024). Mitigating hallucinations is therefore essential for developing reliable AI-supported learning environments (Roychowdhury, 2024). Several mitigation strategies have been proposed. Asbai (2024) examined Retrieval-Augmented Generation (RAG) as a method for reducing hallucinations in ChatGPT and Gemini, evaluating both models on 100 Kattis programming problems. The study found that RAG increased acceptance rates by 10%, reduced error rates, and improved performance across repeated attempts, demonstrating its effectiveness in programming education. In their research, Feldman et al. (2023) suggested an approach to reducing AI hallucinations. They noticed a significant reduction in overall hallucinations when the user provided context, and question prompts to the GenAI tool. Additionally, they assessed how incorporating tags within contexts drastically influenced model responses and significantly eliminated hallucinations with 98.88% efficacy.

Applying Agent AI to detect and mitigate hallucinations in GenAI, Wang and Duan (2025) categorized hallucinations into three types: HK+, a situation in which the model has relevant knowledge but renders erroneous responses; HK-, resulting from inadequate knowledge; and normal answers, which are factual. To address HK-, they utilized corrective RAG to augment missing knowledge, while HK+ was corrected through a human-in-the-loop method. This technique improved LLM accuracy and reliability. This review synthesizes empirical evidence to examine how GenAI hallucinations influence learners' cognitive load, trust calibration, and reasoning processes, and how these effects shape learning performance and problem-solving outcomes in CSE.

3. Methodology

Synthesizing existing research on GenAI-supported personalized and adaptive learning, AI hallucinations, and problem-solving is crucial and timely, particularly in CSE, where emerging AI tools simultaneously offer instructional benefits and introduce new cognitive and pedagogical challenges. A systematic review provides a consolidated view of existing knowledge, enabling the identification of research priorities and addressing questions that individual studies cannot easily resolve. Following PRISMA guidelines (Page et al., 2021), this review systematically examines the role and impact of GenAI in personalized and adaptive learning within CSE. This research adopts the three phases of systematic review as outlined by (Kitchenham & Charters, 2007); the phases include Planning the Review, Conducting the Review, and Reporting the Review.

3.1. Planning the review

3.1.1. Recognizing the need for the review

GenAI reduces instructors' workload by facilitating the creation of educational content that aligns with individual learners' interests (Pesovski et al., 2024). Mahboob et al. (2024) asserted that GenAI's cross-cultural knowledgeability can support the production of culturally relevant and contextually appropriate educational content, thereby enhancing personalized learning for students from diverse backgrounds. However, the understanding of how these tools have been conceptualized and implemented in CSE to address cultural diversity among learners remains limited. Patel (2024) noted that while GenAI hallucinations pose challenges in producing factual and reliable outputs, they may also have the potential to stimulate creativity in personalized learning. This duality underscores the need to examine whether hallucinations influence learners' cognitive processes and whether they hinder or enhance personalized learning outcomes. Additionally, understanding how sustained reliance on GenAI tools affects computer science students' problem-solving and critical thinking skills is essential. These issues reflect long-standing concerns in CSE, including superficial learning strategies, over-reliance on automated support, uneven learning outcomes across cultural contexts, and persistent challenges in fostering deep problem-solving and critical thinking skills. Collectively, these considerations justify the need for a systematic review to analyze, synthesize, and critically evaluate existing empirical evidence in this domain.

3.1.2. Search strategy

The literature search was carried out using four prominent academic databases: Scopus, Web of Science, IEEE Xplore, and ACM Digital Library. These databases were selected based on their extensive coverage of peer-reviewed literature, domain-specific relevance to CSE, and rigorous indexing standards (Falagas et al., 2008; Luxton-Reilly et al., 2018; Martín-Martín et al., 2021). To ensure the inclusion of relevant papers in the review, keywords were identified, search strings were carefully formulated, and a systematic search process was conducted.

3.1.3. Search string protocol formulation

Following the structured approach used by Prather, Leinonen, et al. (2025), we iteratively developed a search string to ensure comprehensive coverage of studies relevant to our research objectives as shown in Table 1. The search string was designed to include papers aligned with our systematic review focus across four key categories: 1) the domain of this study is computer science education, 2) the study examined the role of GenAI in personalized or adaptive learning, 3) it explored at least one of our core research themes: educational outcomes, pedagogical

Table 1
Keyword categories used to formulate the search string.

Category	Keywords
Domain	"Computer science education" OR "Computing education" OR "CS education" OR "CSEd" OR "CER" OR "Computing students" OR "Computing instructors" OR "CS students" OR "CS instructors" OR "Computer engineering education" OR "Programming education" OR "CS pedagogy" OR "Personalized learning in CS" OR "Introductory programming"
Topic (GenAI)	"Generative AI" OR "Large Language Model" OR "LLM" OR "GPT-4" OR "GPT-3" OR "ChatGPT" OR "Claude" OR "Gemini" OR "Deepseek" OR "AI-powered tutoring" OR "Intelligent tutoring systems" OR "Generative pre-trained transformer" OR "GenAI"
Core Research Themes	*Educational Outcomes*: "Academic performance" OR "Student learning outcomes" OR "Learning outcomes" OR "Knowledge retention" *Pedagogical Strategies*: "Personalized learning" OR "Adaptive learning" *Cognitive Processes and Hallucinations*: "Cognitive processes" OR "Metacognitive" OR "Critical thinking" OR "higher order thinking" OR "Problem-solving skills" OR "AI hallucinations" OR "hallucination in AI" *Cultural Diversity and Inclusivity:* "Cultural diversity" OR "Inclusivity" OR "Equity" OR "Bias" OR "Cross-lingual learning" OR "Language diversity in AI" *AI Reliability, Transparency & Fairness*: "Bias in AI" OR "Explainable AI in education"
Method	"Qualitative" OR "Quantitative" OR "Survey" OR "Interview" OR "Experiment" OR "Mixed-methods" OR "Empirical study"

strategies, cognitive processes and hallucinations, and cultural inclusivity, and 4) it employed empirical research methods.

To construct the search string, relevant keywords were identified by reviewing prior literature on CSE (Prather, Leinonen, et al., 2025), GenAI and personalized learning (Osadchyi et al., 2020; Prather, Leinonen, et al., 2025), student learning outcomes (Namoun & Alshanqiti, 2020), AI hallucinations (Maleki et al., 2024), cognitive and higher order thinking skills (Hamzah et al., 2022), cultural diversity and inclusive education (Iniesto & Bossu, 2023; Salas-Pilco et al., 2022), and research method (Prather, Leinonen, et al., 2025). The final search string combined keywords from each category in Equation (1) using Boolean operators, ensuring that included papers addressed at least one keyword from each category:

$$\text{Search String} = \text{Domain AND Topic AND Core Research Themes AND Method} \tag{1}$$

Here, the 'Domain' category includes keywords describing CSE, 'Topic' captures GenAI-related keywords, 'Core Research Themes' represents the key focus areas such as educational outcomes, pedagogical strategies, cognitive processes and hallucinations, and cultural diversity and inclusivity, and 'Method' specifies the empirical research approaches employed in the included studies, as shown in Table 1.

The databases searched and the search scope executed for each database are shown in Table 2. For Scopus, the search scope included article titles, abstracts, and keywords. For IEEE Xplore and ACM Digital Library, searches were conducted within abstracts. For Web of Science, the Topic field (title, abstract, author keywords, and Keywords Plus) was used.

3.2. Conducting the research

3.2.1. Inclusion and exclusion criteria

To ensure a rigorous selection process, we established the inclusion and exclusion criteria as shown in Table 3, drawing from prior research methodologies (Prather, Denny, et al., 2023, Prather, Leinonen, et al., 2025) in computing education and GenAI related studies. These criteria were carefully refined to align with the objectives of this systematic review. The literature search was limited to studies from January 2021 onward to align with the growing relevance of LLMs in computing education. As noted by Prather, Denny, et al. (2023), this period marks an increasing scholarly focus on LLMs, particularly following the emergence of models like GPT-3 in mid-2020. The chosen timeframe ensures the inclusion of recent and significant research in the field.

3.2.2. Paper selection process

The paper selection process adhered to the PRISMA 2020 guidelines, as illustrated in Fig. 1, and consisted of three main phases: Identification, Screening, and Inclusion. During the identification phase, a comprehensive search was conducted across four databases on March 14, 2025, yielding a total of 303 records. After removing 29 duplicate records, 274 unique records remained for screening.

The screening phase involved a two-stage review process. In the first stage (title and abstract screening), 161 records were excluded due to irrelevance to CSE, lack of focus on GenAI, or misalignment with the review's scope. In the second stage (full-text screening), 53 articles were excluded for being non-empirical, in-progress research, or not targeting higher education. Additionally, two articles could not be retrieved in full text despite repeated attempts. In total, 216 records were excluded. A detailed breakdown of the exclusion reasons is provided in Table 4.

To supplement the database search and address the limited number of studies relevant to RQ3, backward and forward snowballing was conducted on studies that had passed the full-text screening stage. This strategy aimed to uncover additional empirical research that may have been overlooked in database indexing. Snowballing yielded three eligible studies. Furthermore, a manual search using Google Scholar, conducted on June 17, 2025, identified three additional empirical studies aligned with the research questions. In total, 64 studies met all eligibility criteria and were retained for quality assessment.

3.2.3. Study quality assessment criteria

In accordance with the adopted guideline (Kitchenham & Charters, 2007), which mandates quality assessment, this study employed a quality assessment checklist from Normadhi et al. (2019), a similar study on personalized and adaptive learning, after applying the inclusion and exclusion criteria. The quality appraisal of the selected studies was conducted to ensure their relevance and reliability. The three-point Likert scale used in each study's quality assessment checklist is presented in Table 5.

Table 2
Databases and search scope.

Academic database	Search scope
Scopus	Article Titles, Abstract, Keywords
Web of Science (WoS)	Topic (consisting of title, abstract, keywords, and Author's keywords)
IEEE Xplore	Abstract
ACM Digital Library	Abstract

Table 3
Inclusion and exclusion criteria.

Criterion	Inclusion	Exclusion
Relevance to GenAI	The study explicitly focuses on GenAI, LLMs, or related tools (such as: ChatGPT, Copilot, Gemini) in higher education.	Studies that do not involve GenAI or only provide theoretical implications without real-world applications.
Computing Education Context	The study must have direct applicability to computing education: Explicit mention of computing education relevance, or research involving students working on computing-related problems, or use of tools or problems designed for computing education.	Papers focusing on AI applications in general education or other disciplines without clear relevance to computing education.
Empirical Contributions	The paper presents empirical findings (such as experiments, case studies, surveys).	Studies that rely only on expert opinions, conceptual discussions, or opinion pieces without empirical data.
Publication Format and Length	The paper must be at least 4 pages long and be a full research article.	Non-research content such as news articles, magazine articles, and posters were excluded.
Publication Date	The study must be published in or after 2021 to ensure relevance to recent advancements in GenAI.	Papers published before 2021 were excluded.
Language	The paper must be written in English to ensure accessibility and consistency in analysis.	Papers not written in English language.

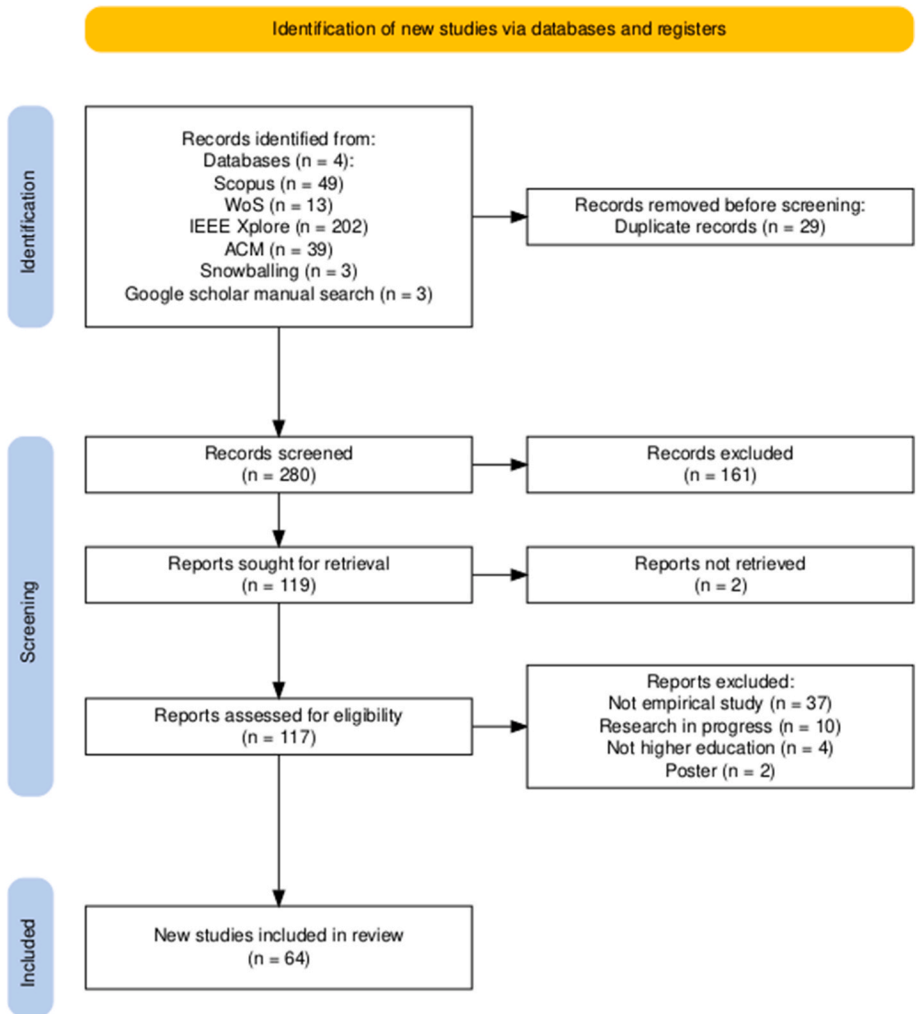


Fig. 1. PRISMA flow diagram.

Table 4
Reasons for excluded records at screening stages.

Screening Stage	Reason for Exclusion	Count (n)
Stage 1	Not related to CS Education	65
	Not focused on GenAI	62
	Not aligned with core themes of the study	29
	Collection of Conference proceedings (not an individual study)	2
	Not written in English (Spanish)	1
	Review papers	2
	Total (Stage 1)	**161**
Stage 2	Not empirical research	37
	Research in progress	10
	Not higher education	4
	Poster	2
	Total (Stage 2)	**53**

Table 5
Quality assessment checklist.

Item	Assessment Criteria	Description of Checklist
QA1	Is the research aim stated?	No. Partially, unclear. Yes, well described.
QA2	Does it discuss core themes?	No. Partially, needs related works. Yes, clearly presented.
QA3	Are the findings described?	No. Partially, unclear details. Yes, comprehensive description.
QA4	Is the methodology stated?	No. Partially, unclear. Yes, clearly stated.
QA5	Is the study cited?	No. Partially, cited by 1–5 articles. Yes, cited by 5+ articles.

3.2.4. Findings of quality assessment

The results of the quality assessment are presented in Fig. 2. In QA1, the research aim of each study was assessed, and all selected studies (100%) clearly described their aims. QA2 evaluated whether the studies aligned with core research themes, revealing that 94% explicitly discussed one or more key themes, while 6% only partially addressed them. For QA3, all studies (100%) provided a clear and detailed description of their findings. Similarly, QA4 confirmed that every study clearly outlined its research methodology. QA5 examined how often the selected studies were cited in other research. Citation counts were used as an indicator of scholarly visibility rather than as an exclusion criterion. We observed inconsistencies in citation counts across different databases. Table 6 presents a few of these inconsistencies. To ensure consistency, we adopted Google Scholar as the primary citation source, following the approach used by.

Normadhi et al. (2019). Interestingly, ACM and Google Scholar had the same citation count for serial number 5. The citation count for each

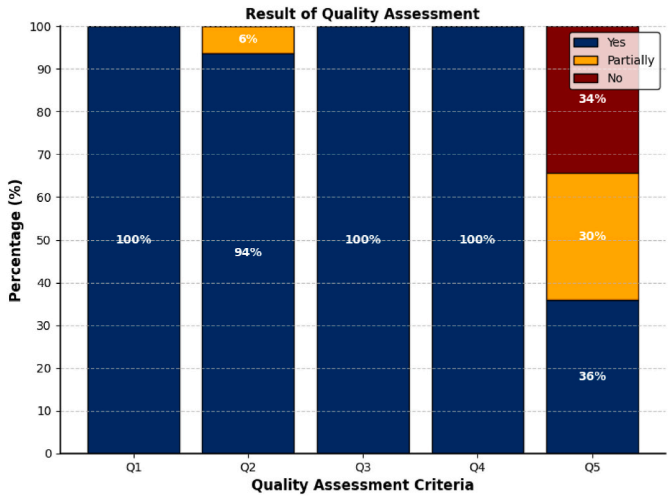


Fig. 2. Result of quality assessment.

Table 6
Citation counts from different databases.

S/N	Title of Paper	Scopus	WoS	IEEE	ACM	Google Scholar
1	Programming education and learner motivation in the age of generative AI: student and educator Perspectives	3	1	-	-	8
2	Challenging the Confirmation Bias: Using ChatGPT as a Virtual Peer for Peer Instruction in Computer Programming Education	8	-	4	-	14
3	Explicitly Introducing ChatGPT into First year programming Practice: Challenges and impact	3	-	1	-	10
4	"It's Weird That it Knows What I Want": Usability and Interactions with Copilot for Novice Programmers	59	36	-	-	185
5	Generative AI in Computing Education: Perspective of Students and Instructors	-	-	19	-	87
6	ChatGPT in the Classroom: An Analysis of Its Strengths and Weaknesses for Solving Undergraduate Computer Science Question	15	-	-	49	49

study was recorded from Google Scholar on March 27, 2025.

Among the 64 studies included, 36% had been cited more than five times, 30% were cited between one and five times, and 34% had not been cited at all. Notably, of the uncited papers, 32% were published in 2025 (the same year this study was conducted), and 64% were published in 2024. These findings highlight that a significant portion of the studies analyzed are recent publications and excluding them solely based on citations may be inappropriate. Hence, this study utilized all 64 papers. The QA5 results may vary over time due to citation updates.

3.3. Data extraction and analysis

The goal of this phase was to systematically capture relevant information from each included study using a structured data extraction approach, guided by the procedures outlined in Kitchenham and Charters (2007). Key data such as study design, educational level, AI tool type, instructional context, and reported outcomes were extracted and organized into a standardized Excel spreadsheet to enable consistent and comparative analysis.

During extraction, entries were coded as "Not addressed" when a study did not engage with a particular variable relevant to the review's research questions. In contrast, the label "Not reported" was applied when a study acknowledged an aspect but failed to provide sufficient detail. This included, for example, studies that referred to ethical protocols without specifying approval procedures, or those that mentioned AI tool usage without clarifying implementation details or personalization mechanisms. These categorizations were essential for identifying both thematic omissions and reporting deficiencies across the evidence base.

4. Results

4.1. Descriptive analysis of the reviewed studies

A total of 64 studies were included in this review. The majority were conference papers (n = 49, 77%), followed by journal articles (n = 14, 22%) and one preprint (1%). The studies were published between 2023 and 2025, with 2024 accounting for the highest number of publications (n = 42), followed by 2025 (n = 13) and 2023 (n = 9). The concentration of studies in 2024 and 2025 reflects the period when GenAI tools became more widely adopted and empirically explored in CSE. Although the review covered studies from 2021 onward, earlier publications were predominantly conceptual or exploratory and did not satisfy the empirical inclusion criteria or align with the review's core themes. This publication trend aligns with the broader trajectory of GenAI integration in education, particularly following the public release of tools such as ChatGPT and GitHub Copilot.

The inclusion of the conference papers is justified because GenAI in CSE is an emerging and rapidly evolving field, where conferences serve as primary dissemination outlets. Many of these conferences (such as ITiCSE, EDUCON, Koli Calling International, ASEE, and FIE) employ rigorous double-blind peer review and often precede journal publications. Following established practice in computing education research (Apiola et al., 2023; Prather, Denny, et al., 2023), this review prioritized top-tier venues. All included studies were screened for methodological rigor, including evaluation design, data sources, validation methods, and peer-review status.

The database search was conducted in March 2025, which may explain the relatively lower number of studies captured for that year due to publication and indexing delays. It is also important to note that some studies address more than one research question; therefore, when reporting the number of studies per research question, the sum may exceed the total number of studies included (64).

4.2. Visualization of emergent themes using word cloud

The word cloud in Fig. 3 visualizes the frequency and prominence of emergent themes identified across the reviewed studies. Larger words represent themes that appeared more frequently, with terms such as "learning," "chatgpt," "student," "AI," and "overreliance" notably emphasized. This visual representation captures the diverse and interconnected nature of the research focus areas, highlighting key concerns around AI hallucinations, critical thinking, prompt engineering, and ethical considerations. The word cloud effectively conveys the thematic framework, aligning with the core research questions of this study and informing our investigation into AI hallucinations and critical thinking.

4.3. Programming languages used in reviewed studies

The analysis of programming languages employed across the reviewed studies as illustrated in Fig. 4, reveals a clear preference for

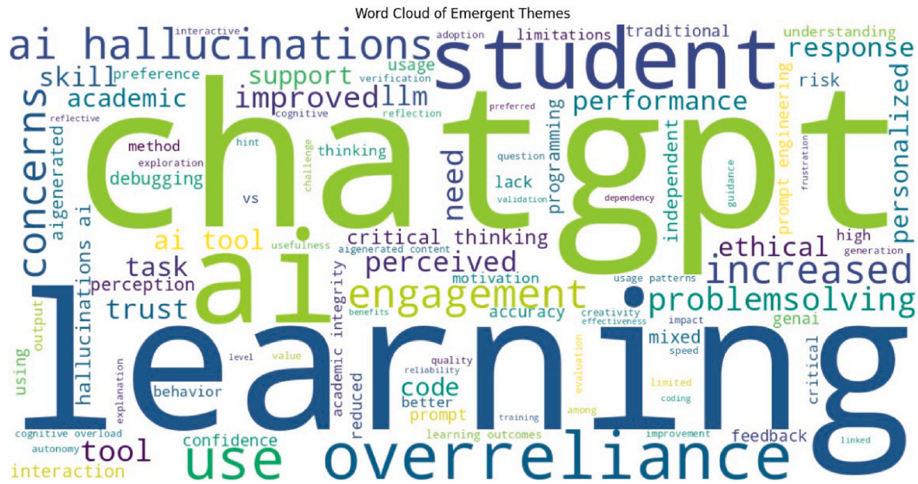


Fig. 3. Word cloud illustrating the most frequent emergent themes across the reviewed studies.

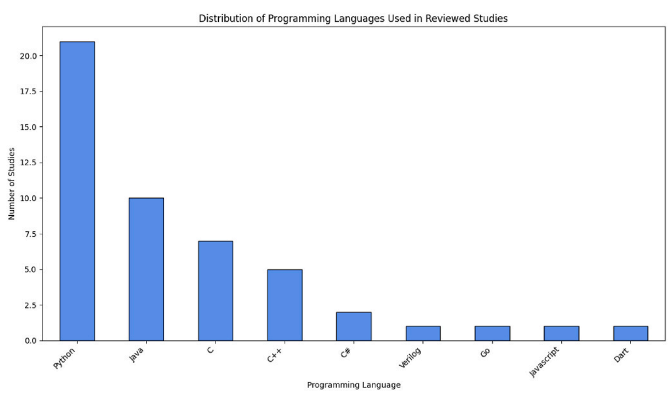


Fig. 4. Distribution of programming languages.

Python, which appears in 21 studies, reflecting its prominence in contemporary CSE and AI integration. Java and C languages also feature prominently, with 10 and 7 studies respectively, indicating their continued relevance in instructional settings. Other languages such as C++, C#, Verilog, Go, JavaScript and Dart are less frequently reported, highlighting a narrower focus on these languages within the current body of research. This distribution underscores the dominant role of versatile, widely adopted programming languages in GenAI educational research.

4.4. Level of study distribution

We analyzed the levels of study reported across the reviewed articles. While some studies focused on a single level (e.g., Undergraduate or Master's), others included multiple levels (e.g., Undergraduate and Master's, or Undergraduate, Master's, and PhD). To ensure accurate representation, combined entries were decomposed, and each level was counted individually. As shown in Fig. 5, undergraduate participants were the most frequently represented (n = 59), followed by Master's (n = 12), while PhD, Postdoctoral, and Diploma levels were each reported in only a few studies. This indicates a strong research emphasis on foundational-level learners and reveals a notable gap in studies targeting more advanced academic levels.

4.5. Study type distribution in reviewed research

The systematic review encompasses a diverse range of study designs employed across the included studies, illustrating the methodological richness in exploring GenAI applications in CSE. As depicted in Fig. 6, the msajority of studies 36 (56%) adopted a Mixed Methods approach, reflecting a balanced integration of qualitative and quantitative data to provide comprehensive insights. Experimental designs constituted the second most common category, 14 (22%), indicating a strong emphasis on controlled empirical investigation.

Qualitative studies accounted for 8 (13%) of the sample, demonstrating the field's interest in in-depth, contextualized understanding of learner experiences and instructional practices. Quantitative studies formed a smaller portion of 6 (9%), typically focusing on statistical

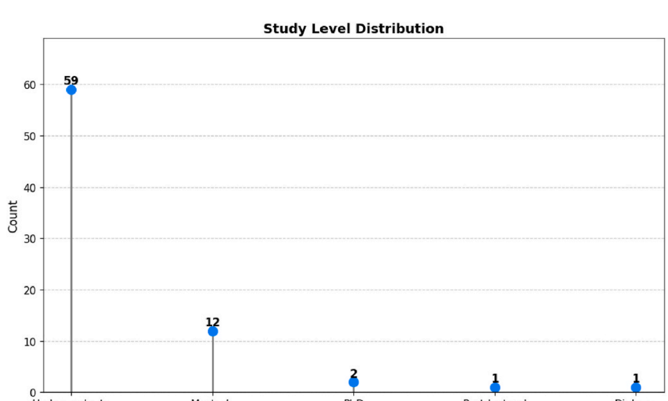


Fig. 5. Level of study distribution of participants across reviewed studies.

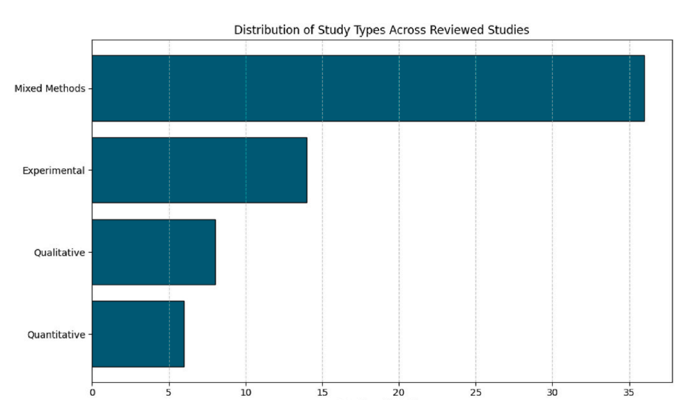


Fig. 6. Distribution of study types across reviewed studies.

analysis of survey data or performance metrics. This distribution highlights the field's inclination to employ multi-faceted research designs to capture the complexity of AI integration in education, while also suggesting opportunities for further purely quantitative or survey-focused research.

The results are organized to reflect a layered view of GenAI-supported learning in CSE: from academic performance outcomes (RQ1), through underlying cognitive influence of hallucinations (RQ2), to contextual and cultural moderators of learning (RQ3), and finally to higher-order problem-solving and critical thinking outcomes (RQ4).

4.6. RQ1: what are the impacts of GenAI-Based personalized learning tools on academic performance in computer science education?

Studies were reviewed to explore the academic impact of GenAI based personalized learning tools in CSE. These studies span diverse contexts, including undergraduate and postgraduate programming courses, lab-based assessments, and project-based learning. A wide range of AI tools were investigated, such as ChatGPT (various versions), GitHub Copilot, Google Gemini, Claude, and several custom or integrated systems (such as CodeTutor, UNCode, Gurukul). Most studies adopted quantitative or mixed-methods approaches, while a few employed qualitative designs or perception-based assessments. The tools were generally used to assist with code generation, debugging, concept clarification, feedback provision, exam preparation, and research planning.

4.6.1. GenAI personalization strategies and academic outcomes

Studies were grouped by the nature of their personalization strategies, as shown in Table 7. The predominant category, user-Driven personalization, involved learners actively shaped their interaction with GenAI tools. This was often achieved through prompt refinement, selection of query types, or iterative engagement based on personal learning needs, as seen in studies such as Joshi et al. (2024); Kusam et al. (2024); Deshpande and Szefer (2023). This approach offered flexibility but yielded varied results depending on the learner's skill level and clarity in input. When learners provided clear, focused inputs, the AI responses were more relevant, but inconsistencies arose in cases lacking clear direction.

The second category, System-Driven (Adaptive) personalization, featured tools that automatically adjusted support based on learner behavior or task progression (Browning et al., 2024; Hasan et al., 2024; Lai & Lin, 2025). These systems required less learner input and produced more consistent positive outcomes, aligning well with pedagogical goals, particularly in structured instructional settings.

As seen in the following thematic analysis, system-driven personalization consistently resulted in stronger academic gains, while user-driven approaches showed a wider range of outcomes. Personalization's effectiveness depends on thoughtful design, system responsiveness, and the extent of instructional scaffolding.

Table 7
Studies grouped by personalization strategy.

Personalization Strategy	Count	Reference
System-Driven (Adaptive)	18	(Amaya et al., 2023; Browning et al., 2024; Chang & Chien, 2024; Hasan et al., 2024; Hayat et al., 2024; Hou et al., 2024; Kazemitabaar et al., 2024; Lai & Lin, 2025; Liao et al., 2024; Liu, Yu, et al., 2024; Lyu et al., 2024; Maurat et al., 2024; Rachha & Seyam, 2024; Shin et al., 2024; Sinha et al., 2024; Winarta et al., 2024; Yeh et al., 2025; Yu et al., 2025)
User-Driven	29	(Abdulla et al., 2024; Axelsson et al., 2024; Baláž et al., 2024; Boguslawski et al., 2025; Brender et al., 2024; Cubillos et al., 2025; de Kereki & Garrido, 2024; Denny et al., 2024; Deshpande & Szefer, 2023; Dos Santos & Cury, 2023; Fenu et al., 2024; Gabbay & Cohen, 2024; Gabriella et al., 2024; Gorson Benario et al., 2025; Hu et al., 2023; Husain, 2024; Joshi et al., 2024; Jošt et al., 2024; Kaleemunnisa et al., 2024; Kusam et al., 2024; Liu, 2024; Park & Kim, 2025; Rahman & Watanobe, 2023; Ramirez Osorio et al., 2025; Schefer-Wenzl et al., 2024; Suárez-Pizzarello et al., 2024; Weber et al., 2024; Xue et al., 2024; Yang et al., 2024)

4.6.2. Adaptive GenAI personalization and consistent academic gains

The integration of adaptive GenAI tools into CSE has demonstrated consistent academic benefits, particularly when systems are designed to personalize feedback, regulate task difficulty, and contextualize instructional support. Across the reviewed studies, a clear pattern emerges: these tools are most effective when aligned with pedagogical principles such as scaffolding, formative feedback, and metacognitive prompting.

A recurring theme involves task-specific feedback in programming education, as shown in studies such as Lai and Lin (2025); Kazemitabaar et al. (2024); Lyu et al. (2024); Yeh et al. (2025). In these cases, personalized guidance targeted students' specific problem areas, leading to measurable gains in accuracy, confidence, and overall performance. Whether delivered through GPT-4-powered tutoring modes or structured hinting systems like CodeAid and CodeTutor, feedback was most effective when embedded in authentic learning contexts and designed to stimulate reflection rather than provide direct answers.

Another observed pattern relates to adaptive difficulty regulation, highlighted in Hou et al. (2024); Amaya et al. (2023); Chang and Chien (2024). These tools dynamically adjusted challenges based on student performance, helping maintain cognitive engagement within the learner's zone of proximal development. Students not only preferred these adaptive systems but also demonstrated stronger conceptual retention, indicating the value of challenge calibration for deeper learning.

Evidence from Kusam et al. (2024); Winarta et al. (2024); Liu, Yu, et al. (2024); Yu et al. (2025) demonstrates how retrieval-augmented generation (RAG) enhanced personalization by grounding AI outputs in course-specific materials. This approach improved response relevance, reduced hallucinations, and fostered learner trust. By fine-tuning outputs using domain-aligned content and enabling iterative prompt validation, RAG-supported tools promoted instructional coherence and more meaningful engagement with academic material.

Despite these benefits, adaptive personalization is not without risks. Liao et al. (2024) cautioned that while computational thinking skills improved, independent problem-solving declined when students became overly reliant on AI. This illustrates a key design challenge: while GenAI tools can scaffold understanding, they must also support learner autonomy to avoid eroding self-directed reasoning.

The findings suggest that the academic benefits of adaptive personalization depend on specific design features, real-time feedback, challenge calibration, and contextual grounding. When these elements are effectively implemented, GenAI tools serve as powerful cognitive companions that support student performance, conceptual development, and reflective problem-solving. In the absence of such features, even well-intentioned personalization may lead to passive engagement or overdependence. This aligns with the adaptive learning design principles articulated by Cavanagh et al. (2020), who emphasize that effective AI-driven personalization requires the integration of real-time adaptation, cognitive scaffolding, learner engagement strategies, and transparent instructional logic to optimize learning outcomes while safeguarding against systemic risks.

4.6.3. Self-directed personalization

The synthesized studies revealed how learner-led personalization, where students independently shaped GenAI interactions through prompt crafting, iteration, or selective engagement yielded highly variable yet academically relevant outcomes. Unlike adaptive systems that dynamically tailor responses based on performance data, user-driven personalization relies heavily on the learner's ability to formulate meaningful prompts, manage their interaction, and extract relevant output. While this approach offers considerable flexibility and learner agency, the academic benefits observed were markedly inconsistent, influenced by differences in learner expertise, instructional context, and the complexity of the AI interface.

A recurring pattern emerged around prompt engineering and user engagement (Hu et al., 2023; Joshi et al., 2024; Kaleemunnisa et al., 2024; Yang et al., 2024). These studies demonstrated that when learners framed precise, task-specific prompts, they experienced improvements in problem-solving and task accuracy. However, these benefits varied with topic complexity and learner proficiency. In several cases, excessive reliance, particularly among novice learners, led to reduced independent reasoning.

Studies such as Gabbay and Cohen (2024); Gabriella et al. (2024); Shin et al. (2024) highlighted motivational gains, where personalized AI feedback helped reduce frustration, increase engagement, and support sustained task completion. Although direct academic metrics were not always reported, learners' confidence and perceived understanding improved, especially when they actively iterated with AI systems to refine their thinking. Cautionary evidence from Jošt et al. (2024); Baláž et al. (2024) showed that over-reliance on AI tools could lead to diminished academic performance and reduced depth of understanding. These findings suggest that efficiency-focused use of GenAI may bypass critical thinking, resulting in shallow, convergent solutions.

These patterns underscore that learner-led personalization can yield motivational and academic benefits but only when supported by sufficient learner skill, reflective habits, and instructional framing. Without such scaffolding, personalization risks reinforcing surface-level engagement rather than promoting deep learning.

Table 8
Categorization of studies by impact outcome.

Impact Category	Count	Reference	Description
Positive	19	(Abdulla et al., 2024; Browning et al., 2024; Chang & Chien, 2024; de Kereki & Garrido, 2024; Denny et al., 2024; Dos Santos & Cury, 2023; Gabriella et al., 2024; Gorson Benario et al., 2025; Hasan et al., 2024; Hou et al., 2024; Kazemitabaar et al., 2024; Kusam et al., 2024; Liu, 2024; Maurat et al., 2024; Park & Kim, 2025; Rachha & Seyam, 2024; Shin et al., 2024; Yang et al., 2024; Yeh et al., 2025)	The use of GenAI based personalized tools resulted in measurable academic improvement, such as higher grades, better task performance, and more efficient learning.
Negative	2	(Jošt et al., 2024; Ramirez Osorio et al., 2025)	Academic performance declined due to AI use, often caused by over-reliance, confusion, and misleading out-puts.
Mixed	12	(Baláž et al., 2024; Brender et al., 2024; Deshpande & Szefer, 2023; Gabbay & Cohen, 2024; Hu et al., 2023; Joshi et al., 2024; Lai & Lin, 2025; Liao et al., 2024; Lyu et al., 2024; Rahman & Watanobe, 2023; Schefer-Wenzl et al., 2024; Suárez-Pizzarello et al., 2024)	The AI tool helped in some areas but hindered in others, with outcomes varying by task type, student experience, and implementation strategy.
Neutral	2	(Cubillos et al., 2025; Xue et al., 2024)	No significant change in academic performance was observed despite the use of GenAI tools.

4.6.4. Distribution of academic outcomes from GenAI personalization

To better understand the impact of GenAI-based personalized learning tools on academic performance in CSE, the reviewed studies were classified into four outcome categories: Positive, Negative, Mixed, and Neutral (Table 8). Out of the 35 studies that explicitly reported academic performance outcomes, 19 (54.3%) reported positive academic impacts, including measurable gains in test scores, task completion accuracy, learning speed, and pass rates. Twelve studies (34.3%) demonstrated mixed effects, where tools supported some tasks or learners but hindered others due to variations in tool behavior, student background, or prompt formulation quality. Negative outcomes were reported in 2 studies (5.7%), primarily linked to over-reliance on AI for critical thinking-intensive tasks such as code generation or debugging, leading to lower grades and reduced problem-solving autonomy. Two studies (5.7%) found no significant effect on academic performance.

Fig. 7 visualizes the proportional distribution of these outcomes, highlighting that over half of these studies indicate a positive influence of GenAI personalization, particularly when tools incorporate adaptive or scaffolded support. Mixed outcomes were more common in user-driven personalization settings, where the effectiveness depends on learners' skill in prompt engineering and engagement strategies. Negative or neutral impacts, though infrequent, emphasize the potential risks of unguided AI use.

This outcome-based synthesis suggests that well-structured and guided personalization, especially adaptive or instructor-mediated approaches, consistently promotes academic benefits. In contrast, loosely implemented or unguided interactions may yield inconsistent or negligible improvements, underscoring that thoughtful pedagogical design is as crucial as the GenAI tools themselves in achieving meaningful learning outcomes.

Across the reviewed studies, academic performance outcomes were consistently associated with learners' cognitive regulation when interacting with GenAI tools. Performance gains were reported in contexts where learners monitored AI feedback, calibrated reliance, and strategically regulated their engagement, whereas performance declines were observed in studies reporting over-reliance or limited regulatory control.

4.7. RQ2: how do GenAI hallucinations influence cognitive processes and learning outcomes in computer science education?

This section examines how hallucinations from GenAI tools influence cognitive processes and learning outcomes in CSE. Hallucinations, defined as plausible but incorrect outputs, can disrupt reasoning or, in some contexts, prompt critical reflection. Studies were analyzed based on how learners interacted with these outputs, with attention to error detection, cognitive regulation, and task progression. Two primary

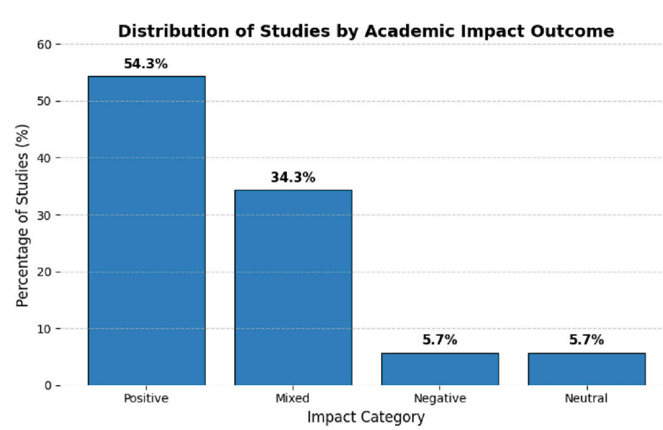


Fig. 7. Proportion of studies by reported academic impact outcome.

categories emerged: Cognitive Disruption and Metacognitive Activation. These categories capture the varied ways learners engage with hallucinated content, shaped by factors such as prior knowledge, task complexity, and instructional design. This classification highlights both the pedagogical risks and the potential learning opportunities associated with AI hallucinations in educational settings.

4.7.1. *Cognitive disruption: the pedagogical cost of AI hallucinations*

Cognitive disruption refers to the challenges students encounter when hallucinated or inaccurate outputs interfere with reasoning, critical thinking, and engagement with learning tasks. These hallucinations often lead to confusion, increased cognitive load, and over-reliance on AI-generated content, ultimately hindering students' ability to process feedback, construct understanding, and develop problem-solving skills. A consistent pattern across studies is the substantial cognitive strain caused by hallucinations, particularly in precision-dependent tasks such as programming, numerical reasoning, and abstract logic. Hallucinated outputs are frequently fluent yet incorrect, leading students to accept them as accurate due to the AI's confident presentation. For example, Joshi et al. (2024); Deshpande and Szefer (2023); Winarta et al. (2024); de Kereki and Garrido (2024) document how fabricated diagrams, faulty logic, and incorrect code misled students into trusting flawed responses. The absence of uncertainty cues in AI outputs further exacerbated this problem, as students often failed to verify responses and instead relied on the system's apparent authority.

Over-reliance on AI tools, especially among novice learners, emerged as a major contributor to cognitive disruption. Studies by Schefer-Wenzl et al. (2024); Khan et al. (2023); Wei et al. (2023) show that many students accepted hallucinated outputs without scrutiny, bypassing critical evaluation processes. This dependence undermines the development of independent reasoning and problem-solving skills, as learners substitute AI authority for their own analytical engagement.

In addition, cognitive overload is a recurrent theme. Lai and Lin (2025), Lyu et al. (2024), Yu et al. (2025), Hou et al. (2024) illustrate how vague, overly complex, or erroneous AI outputs contribute to cognitive strain. These unclear or misleading responses, such as ambiguous feedback or irrelevant content, increase cognitive load and hinder students' ability to process information effectively. In some cases, students experienced frustration or disengagement, particularly when AI responses did not align with their expectations or learning objectives, thus impairing their ability to continue focusing on the task at hand.

The challenges of cognitive disruption are especially pronounced in programming contexts, as highlighted in Husain (2024); Ramirez Osorio et al. (2025); Prather et al. (2024); Hou et al. (2024); Lyu et al. (2024). Hallucinated AI-generated code, referred to as code hallucination, often misguides students, leading to confusion about key programming concepts. Misunderstandings of code logic and structure can distort students' mental models, resulting in superficial engagement with the material. The tendency to blindly copy AI-generated code, as observed in these studies, without understanding the underlying logic, reinforces misconceptions and impedes deeper learning and skill development in programming.

These findings illustrate the alignment with cognitive load theory (Elsayed, 2024), specifically, the overwhelming or misleading information caused by AI hallucinations can overload working memory and obstruct deeper cognitive engagement. The misleading information, presented confidently by AI, disrupts the integration of new content into existing cognitive models, as suggested by Constructivist theories of learning (Harris et al., 2013). This over-reliance on AI prevents critical engagement, impeding students' ability to construct accurate mental models. To mitigate these effects, interventions such as AI literacy training, task verification strategies, and educator involvement are essential to ensure GenAI tools support learning effectively.

4.7.2. *Hallucinations as catalysts for metacognitive engagement*

Inaccurate or AI hallucinations from GenAI tools can act as catalysts for metacognitive engagement, prompting students to reflect, self-monitor, and refine their cognitive strategies. Across several studies, students did not passively accept flawed outputs; instead, they engaged with increased skepticism and cognitive vigilance. For example, in studies such as Kusam et al. (2024); Hasan et al. (2024); Suárez-Pizzarello et al. (2024) and Prather, Reeves, et al. (2023), students refined prompts, validated responses, and adjusted their approaches, demonstrating key elements of metacognitive awareness such as self-regulation and active monitoring. Although occasional confusion was reported, these interactions generally enhanced deeper engagement and reflective learning practices.

Instructor support and system design also played a pivotal role in guiding students' metacognitive development (Dos Santos & Cury, 2023; Yeh et al., 2025). In the mentioned studies, students initially placed undue trust in AI outputs, but educator intervention or interface designs that revealed uncertainty helped them recalibrate their expectations. These guided experiences not only reduced blind reliance but also reinforced the view of GenAI as a tool for exploration rather than a definitive authority. Other studies, including Kazemitabaar et al. (2024); Axelsson et al. (2024); Sinha et al. (2024), illustrated how AI hallucinations prompted students to adjust their trust in GenAI systems. While novice users were particularly susceptible to confidently incorrect responses, many became more cautious and reflective over time. Even flawed outputs were frequently repurposed as opportunities for alternative thinking and deeper conceptual exploration, highlighting their potential to support higher-order cognitive skills when scaffolded by appropriate awareness and guidance.

These studies reveal that hallucinations, while commonly seen as obstacles, can become productive triggers for metacognitive growth when learners are supported in navigating uncertainty. Whether through deliberate pedagogical scaffolding, reflective tool design, or the inherent limitations of AI, students in these settings developed habits of inquiry, prompt refinement, and critical assessment. These findings align with the emerging view that the cognitive risks of hallucinations can be rechanneled into cognitive assets when students are prompted to interrogate, rather than accept AI outputs (Drosos et al., 2025; Tankelevitch et al., 2024).

4.7.3. *Quantitative summary of cognitive influence*

Fig. 8 presents a summary of the 43 studies classified by their reported cognitive influence. Most studies (25) exhibited mixed effects, indicating that hallucinations can both support and hinder learning depending on context. Nine (9) studies reported purely hindering effects, reflecting situations where students were misled by confidently incorrect AI outputs, leading to cognitive disruption, over-reliance, or reduced problem-solving performance. Another nine (9) studies

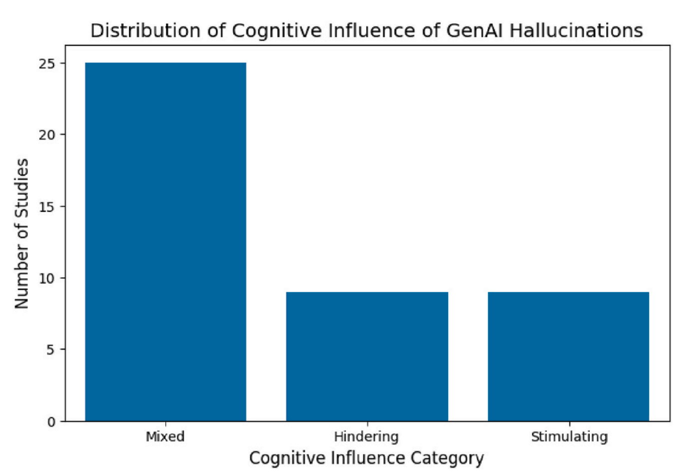


Fig. 8. Distribution of studies by cognitive influence of GenAI hallucinations.

reported stimulating effects, where hallucinations prompted metacognitive engagement, encouraging students to critically evaluate, reflect, and refine their reasoning. These quantitative patterns align with the qualitative observations discussed earlier, highlighting that the impact of hallucinations is nuanced and influenced by factors such as prior knowledge, instructional support, and task complexity.

These findings indicate that the impact of GenAI hallucinations is contingent on learners' cognitive regulation. Hallucinations disrupted learning in contexts where students failed to monitor accuracy or calibrate trust, whereas they stimulated learning when they prompted reflective evaluation, error detection, and strategic adjustment.

4.8. RQ3: how do GenAI platforms support culturally diverse learners' cognitive engagement in computer science education?

As GenAI platforms become increasingly embedded in CSE, questions of equity and contextual relevance have gained prominence. This research question examines how these technologies support learners from culturally diverse backgrounds and what strategies are used to promote contextual inclusivity. Given the global and multilingual nature of computer science classrooms, it is essential to examine not only the design of GenAI tools but also their accessibility, cultural adaptability, and responsiveness to diverse learners' needs. However, empirical research at the intersection of GenAI, cultural inclusivity, and CSE remains limited. Much of the existing literature is conceptual or technology-centered, with relatively few studies examining how GenAI operates across diverse learner populations in authentic Computer Science learning contexts. Accordingly, the findings in this review are organized around three themes: language inclusivity, cultural contextualization, and equity in personalization and access. In this review, cultural diversity is examined as interacting with learner's cognitive processes, shaping how they engage with and benefit from GenAI-supported learning rather than functioning as an independent instructional mechanism.

4.8.1. Language inclusivity

Studies indicate that GenAI platforms can support linguistically diverse learners by enabling multilingual interaction (Jordan et al., 2024; Prather, Reeves, et al., 2025). These studies explore how multilingual prompting, allowing learners to interact with GenAI in native languages such as Arabic, Chinese, Spanish, Vietnamese, and Tamil, can alleviate language-based barriers in CSE. The evidence suggests that native-language input enables deeper engagement and reduces intimidation among non-native English speakers, thereby enhancing an inclusive environment for students historically marginalized by English-dominant curricula. However, multilingual effectiveness is uneven. While high-resource languages benefit from relatively coherent and functional GenAI outputs, low-resource languages (e.g., Tamil) reveal substantial limitations, including syntactic and semantic inconsistencies. These findings affirm that while GenAI offers potential to make language access more equitable, it also exposes persistent imbalances, since many languages are still poorly supported due to limited representation in the data used to train these models. Consequently, GenAI-facilitated language inclusion will require targeted improvements in multilingual modeling and equitable support across languages.

4.8.2. Cultural contextualization

GenAI-generated programming exercises that reflect learners' interests and local contexts have been shown to enhance motivation, engagement, and perceived relevance (del Carpio Gutierrez et al., 2024; Oyelere & Aruleba, 2025). These studies highlight the potential of GenAI to personalize content in ways that make programming tasks more meaningful, particularly for students from diverse cultural and socio-economic backgrounds. Such contextualization can support inclusivity by helping learners feel more connected to course material. For example, AfriML, developed by Okafor et al. (2025), integrates African linguistic and cultural elements, improving engagement and accessibility in machine learning education. Similarly, Oyelere et al. (2022) demonstrate how AI tools tailored to local cultural contexts in African schools improve student engagement, underscoring the importance of contextualized resources in educational tools.

When learners could relate to task contexts, both engagement and performance improved. In contrast, when cultural references were unfamiliar, superficial, or misaligned, tasks introduced unnecessary cognitive load and reduced learning effectiveness. Although GenAI presents opportunities for culturally responsive support, inconsistencies in the representation of diverse perspectives highlight the need for pedagogically grounded and culturally adaptive design, rather than surface-level localization.

4.8.3. Equity in personalization and access

Recent studies indicate that GenAI-powered learning platforms often reflect pre-existing educational inequalities, despite their scalable design. The benefits of personalization are not equitably distributed, with outcomes shaped by socioeconomic status (SES), gender, and infrastructural access (Maurat et al., 2024; Pang et al., 2024; Prather, Reeves, et al., 2025).

Students from low-income or rural backgrounds frequently use GenAI tools for foundational support, such as comprehension assistance and basic debugging (Pang et al., 2024). In contrast, more privileged students tend to employ these tools for advanced problem-solving, optimization, and exploration. This dual usage pattern suggests that GenAI often functions as a compensatory aid for disadvantaged learners but as a performance amplifier for others, thereby reinforcing, rather than reducing, academic stratification in CS education.

Gender-based disparities further complicate equitable access. Although female students report similar or greater perceived usefulness of GenAI tools, they engage less frequently in AI-supported programming tasks (Maurat et al., 2024). This reduced engagement is often linked to lower confidence and limited familiarity with AI-assisted workflows. As such, equitable personalization in CS education must extend beyond adaptive algorithms to include pedagogical scaffolds that build learner confidence and normalize sustained engagement across underrepresented groups.

Structural barriers remain a significant concern. Students in under-resourced contexts face limitation such as device incompatibility, paywalled platforms, and unstable internet access, which restrict meaningful interaction with GenAI systems. According to Pang et al. (2024); Oyelere and Aruleba (2025), without institutional interventions, such as subsidized deployments, regionally optimized tools, and mobile-compatible access, GenAI risks exacerbating the digital divide rather than closing it. These inclusivity challenges are summarized in Table 9, which outlines how equity concerns manifest across personalization, gender engagement, and infrastructural access.

4.8.4. Quantitative summary of inclusivity dimensions

Across the six included studies for RQ3, inclusivity was examined from three primary perspectives: language inclusivity, cultural contextualization, and equity in personalization and access. Two studies explicitly addressed language inclusivity (Jordan et al., 2024; Prather, Reeves, et al., 2025), two focused on *cultural contextualization* (del Carpio Gutierrez et al., 2024; Oyelere & Aruleba, 2025), and three examined *equity in personalization and access* (Maurat et al., 2024; Pang et al., 2024; Prather, Reeves, et al., 2025). Notably, one of the studies (Prather, Reeves, et al., 2025) addressed more than one dimension, covering both language and equity aspects. This distribution indicates that while there is emerging empirical attention to inclusivity in GenAI-supported CS education, coverage remains limited and uneven, particularly for language and cultural contextualization, highlighting the need for broader and more systematic empirical work. See Table 9 for a detailed summary of inclusivity evidence across the included studies.

Table 9
Inclusivity dimensions in GenAI-Supported computer science education.

Inclusivity Dimension	Focus Area	Key Findings	Recommended Strategies
Language Inclusivity (Jordan et al., 2024; Prather, Reeves, et al., 2025)	Multilingual prompting in CS education	Native-language input enables learners to engage more deeply in programming tasks such as debugging and code explanation. However, learners using low-resource languages (e.g., Tamil, Arabic) face reduced accuracy and coherence due to limited model training data.	Improve LLM performance for underrepresented languages; expand multilingual support for common programming use cases in CS education.
Cultural Contextualization (del Carpio Gutierrez et al., 2024; Oyelere & Aruleba, 2025)	Culturally relevant GenAI-generated programming scenarios	Students perform better and show greater motivation when programming tasks reflect familiar cultural or socioeconomic contexts. Poorly contextualized or superficial examples introduce cognitive load and reduce engagement.	Embed locally relevant cultural themes into problem design; employ culturally grounded pedagogy when generating personalized content in GenAI-supported platforms.
Equity in Personalization and Access (Maurat et al., 2024; Pang et al., 2024; Prather, Reeves, et al., 2025)	Variation in GenAI use by SES, gender, and infrastructure	Learners from lower-SES backgrounds tend to use GenAI for basic comprehension and debugging, while more privileged peers use it for optimization and exploration. Female students re-port lower confidence and engagement. Access disparities persist due to device limitations, paywalls, and internet instability.	Promote bias-aware personalization, design scaffolds to boost tool confidence among underrepresented groups, and implement mobile-optimized, low-cost deployment strategies to bridge infrastructure gaps.

This distribution aligns with the methodological diversity observed across the six studies. The six studies examining inclusivity in GenAI-supported CS education employed a mix of experimental, survey-based, and mixed-methods designs. Two studies (Jordan et al., 2024; Prather, Reeves, et al., 2025) evaluated LLM outputs and native-language prompts, with Prather, Reeves, et al. (2025) combining quantitative analysis of model outputs with student survey feedback, and Jordan et al. (2024) using automated code testing alongside manual assessment by native speakers to evaluate linguistic and contextual quality. Two studies (del Carpio Gutierrez et al., 2024; Oyelere & Aruleba, 2025) investigated culturally contextualized programming exercises, with Oyelere and Aruleba (2025) conducting a mixed-methods survey of undergraduate students across Sub-Saharan Africa, and del Carpio Gutierrez et al. (2024) performing an experimental study to evaluate GPT-4–generated exercises with varied prompting strategies and culturally diverse contexts.

Maurat et al. (2024) conducted a large-scale survey of 342+ undergraduate students to examine gender differences in AI tool use and perceived efficacy, while Pang et al. (2024) combined surveys with semi-structured interviews to explore socioeconomic influences on AI adoption. Across these studies, common features include undergraduate student samples, context-specific tasks, and validated instruments such as Likert scales or rubrics. Overall, the body of work demonstrates a predominance of quantitative assessments complemented by qualitative insights, highlighting methodological diversity in studying language, culture, and equity in GenAI-supported CS education.

Across the reviewed studies, cultural and linguistic diversity functioned as a contextual moderator of cognitive engagement rather than as an independent learning outcome. Learners' familiarity with AI-generated content influenced how they interpreted, evaluated and applied GenAI-supported guidance, shaping downstream effects on engagement, learning, and problem-solving in CSE.

4.9. Theoretical foundation of adaptive GenAI in problem-solving and critical thinking (RQ4)

RQ4 examines how adaptive GenAI-driven tools influence the development of problem-solving and critical thinking skills in CSE. Programming contexts are characterized by high algorithmic complexity (Huang et al., 2025), intensive debugging demands (Hoq et al., 2025), uneven prior knowledge (Qi et al., 2025), and increasing reliance on automated assistance (Feng et al., 2025). Under these conditions, higher-order reasoning depends on adaptive scaffolding that shapes learners' cognitive engagement (Borge et al., 2024; Lee et al., 2025), aligning with recent categorizations of GenAI's influence across learning and competency dimensions (Hazzan & Erez, 2025). RQ4 therefore draws on Constructivist Learning Theory, Cognitive Load Theory, Adaptive Learning Theory, and Metacognitive Learning Theory to clarify the mechanisms through which adaptivity influences higher-order skill development. From a constructivist perspective, programming expertise develops through active abstraction, iterative reasoning, and conceptual reconstruction (Hsu, 2025; Matobobo et al., 2025). GenAI can scaffold these processes through explanation and decomposition; however, when adaptive systems bypass productive struggle, they may inhibit the development of independent reasoning (Kohen-Vacs et al., 2025; Matobobo et al., 2025).

These constructive processes interact closely with cognitive load, as the mental effort required for abstraction and debugging influences how learners engage with GenAI scaffolding. Cognitive Load Theory is particularly relevant in programming, where tasks impose high intrinsic load (Huang et al., 2025; Nagori et al., 2025). Adaptive feedback can support critical thinking by reducing extraneous cognitive burden and enabling deeper strategic processing (Huang et al., 2025). However, excessive assistance may diminish engagement in effortful reasoning (Nagori et al., 2025).

The regulation of cognitive load is further shaped by system-level adaptivity, which calibrates task difficulty and feedback to sustain optimal engagement. Adaptive Learning Theory explains how dynamic calibration of difficulty and feedback sustains iterative reasoning within an optimal challenge range (Hazzan & Erez, 2025; Tarun et al., 2025). When instructionally aligned, such adaptivity strengthens strategic problem-solving; when efficiency-driven, it risks fostering dependency.

Although adaptivity regulates task challenge, critical thinking relies on learners' metacognitive strategies to evaluate and regulate their engagement with AI outputs. Metacognitive Learning Theory clarifies that critical thinking requires monitoring, verification, and strategic adjustment (Sultana & Sianaki, 2025), especially when learners encounter ambiguous or partially incorrect AI outputs (Elsayed, 2024). Adaptive systems that prompt reflection can strengthen regulatory control, whereas unguided use may externalize evaluation to the system (Borge et al., 2024; Hazzan & Erez, 2025). These theories explain how adaptive GenAI tools shape the development of problem-solving and critical thinking skills through their interaction with cognitive load, constructive engagement, and metacognitive regulation.

4.10. RQ4: how do adaptive learning tools driven by GenAI influence problem-solving and critical thinking skills in computer science students?

This synthesis examines the influence of adaptive learning tools driven by GenAI on the development of problem-solving and critical thinking skills among computer science students. Drawing upon a selected body of empirical studies, the analysis considers the cognitive affordances and potential drawbacks associated with GenAI integration. Particular attention is given to the context of tool deployment, instructional design features, patterns of learner engagement, and documented learning outcomes. Providing an evidence-based account of the extent to which these tools function as cognitive enhancers, scaffolding mechanisms, or, conversely, as sources of dependency that may inhibit the cultivation of higher-order thinking skills in CSE.

Building on findings from RQ2, hallucinations in GenAI outputs can both hinder and stimulate students' problem-solving and critical thinking. While cognitive disruption may lead to errors or superficial reasoning, it can also trigger metacognitive activation that encourages reflection, iterative debugging, and deeper strategic thinking during programming tasks.

4.10.1. GenAI tools and their modes of integration

This review synthesizes how GenAI tools were integrated into CSE, focusing on instructional design rather than tool type alone. Tools ranged from general-purpose LLMs such as ChatGPT, Claude, StarCoder, and GitHub Copilot, to custom platforms like BoilerTAI, PyTutor, CodeAid, Gurukul, and QuickTA, often extended with task-specific scaffolding or feedback mechanisms as shown in Table 10.

In several studies (Axelsson et al., 2024; de Kereki & Garrido, 2024; Fenu et al., 2024; Prather et al., 2024; Tayeb et al., 2024), students used GenAI tools independently via public interfaces. These informal uses, typically during assignments or debugging tasks, offered flexibility but often lacked pedagogical scaffolding. This led to surface-level engagement and over-reliance, especially when students lacked prompting skills or conceptual grounding. In contrast, tools like CodeTailor (Hou et al., 2024), PyTutor (Yang et al., 2024), and Gurukul (Rachha & Seyam, 2024)) were explicitly designed to scaffold problem-solving. Features such as structured hinting, adaptive sequencing, and delayed solution exposure supported iterative reasoning and reflective learning.

Other platforms (BoilerTAI (Sinha et al., 2024), CodeAid (Kazemitabaar et al., 2024), IPSSC (Liao et al., 2024)) were embedded into intelligent tutoring systems, enabling real-time monitoring, context-sensitive feedback, and alignment with curricular objectives. Their integration promoted sustained engagement, particularly when instructors mediated AI-student interaction. Some implementations emphasized collaboration, in Dos Santos and Cury (2023); Styve et al. (2024), GenAI acted as a peer reviewer or pair programmer, supporting dialogic reasoning and code refinement. These peer-like roles promoted critical thinking, though their effectiveness hinged on instructor oversight. A few studies (Kusam et al., 2024; Liu, Yu, et al., 2024; Yu et al., 2025) used retrieval-augmented generation (RAG) or small language models (SLMs) to deliver course-specific, context-aware feedback. These tools narrowed the semantic gap between general LLM outputs and domain-relevant reasoning, enhancing precision and conceptual alignment. In performance-driven settings (Shin et al., 2024; Takerngsaksiri et al., 2024, pp. 1606–1611), GenAI tools provided real-time feedback during timed or competitive tasks. These environments emphasized rapid application of problem-solving skills and offered insight into how

Table 10
Overview of AI tool characteristics and integration.

Reference	Type of AI Model Used	Purpose of AI Tool	AI Tool's Integration
Kusam et al. (2024)	ChatGPT (LLM), with finetuning and retrieval-augmented generation (RAG)	To teach prompt engineering and basic GenAI tool customization through project-based learning.	Embedded in an AI course using OpenAI API, Gradio, and Colab for hands-on learning
Sinha et al. (2024)	GPT-4 (OpenAI)	To assist teaching assistants by generating draft responses to programming queries	BoilerTAI integrated with EdDiscussion forums for instructor-reviewed AI replies
Kazemitabaar et al. (2024)	GPT-3.5-turbo; formerly Code-DaVinci	Provide structured hints and feedback to support code understanding and debugging	Integrated into CodeAid as a feedback-driven support system for programming students
Hou et al. (2024)	GPT-4 (via CodeTailor)	Generate adaptive Parsons puzzles to scaffold programming logic and debugging skills	Embedded in CodeTailor with progressive difficulty and reflective prompts for CS students
Yang et al. (2024)	GPT-4 (via PyTutor)	Provide structured hints and guided feedback for solving programming problems	Deployed in PyTutor with restricted output and scaffolded hinting to pro-mote reasoning
Fenu et al. (2024)	ChatGPT (GPT-3.5)	Assist students during informal debugging and writing tasks	Accessed independently via Chat-GPT's public web interface
Axelsson et al. (2024)	ChatGPT (GPT-3.5)	Used informally by computing students for academic support, idea generation, and writing assistance	Accessed independently without formal integration into course structures
Yu et al. (2025)	GPT-3.5 and customized small models (SLMs)	Provide secure, context-aware AI support using retrieval pipelines tailored to course content	Integrated via the Gemini platform with retrieval-augmented generation for private, course-aligned interaction
Rachha and Seyam (2024)	GPT-3.5-turbo (via Gurukul)	Offer hint-based assistance in problem-solving without revealing solutions	Built into Gurukul coding environment with structured feedback flow
Liao et al. (2024)	Claude 2 via IPSSC	Facilitate iterative reasoning in software planning tasks	Embedded in IPSSC tutor for scaffolded design thinking
de Kereki and Garrido (2024)	ChatGPT (GPT-3.5)	Support code generation, debugging, and logic explanation in individual programming tasks	Public interface use encouraged within a structured CS course, with in-class demonstration and prompt guidance
Shin et al. (2024)	GPT-4 via CodeHelp in Codio	Real-time feedback in quiz-based or timed programming activities	Integrated with Codio platform for adaptive challenge delivery
Dos Santos and Cury (2023)	ChatGPT (peer reviewer role)	Support reflective thinking and code review as part of peer instruction	Used in peer-learning model with guided prompting strategies
Prather et al. (2024)	ChatGPT	Support debugging and code reasoning	Used independently by students without

(continued on next page)

Table 10 (*continued*)

Reference	Type of AI Model Used	Purpose of AI Tool	AI Tool's Integration
		in unstructured programming tasks	structured course guidance
Styve et al. (2024)	GitHub Copilot	Serve as a pair programming partner during code writing and revision	Embedded in IDE with reflective review exercises
Takerngsaksiri et al. (2024)	StarCoder (via AutoAurora)	Provide code completion, syntax support, and alternative solutions to assist student coding	Used in a VS Code extension during structured programming tasks with follow-up interviews

students adapt under cognitive pressure.

Integration depth, not model complexity, was the most consistent predictor of educational benefit. Tools embedded into structured, scaffolded contexts supported metacognition and iterative reasoning, while unstructured use often led to passive engagement and automation bias. These findings align with Holstein et al. (2020) framework of human–AI hybrid adaptivity, which positions adaptivity as a collaborative process between AI systems and educators. When GenAI tools are instructionally orchestrated, they function not merely as generative assistants, but as co-participants that promote deeper engagement, critical reasoning, and learner autonomy.

4.10.2. Cognitive impacts on problem-solving and critical thinking

The integration of GenAI tools into CSE has produced mixed cognitive outcomes, with impacts on problem-solving and critical thinking varying according to the structure of use, learner ability, and instructional context. In studies such as Joshi et al. (2024); Kusam et al. (2024); Lai and Lin (2025); Sinha et al. (2024); Liu, Yu, et al. (2024), structured implementations where GenAI tools like ChatGPT and BoilerTAI were embedded into pedagogical frameworks, supported deeper reasoning and adaptability. These tools provided guided assistance and feedback that encouraged reflection rather than passive answer retrieval, resulting in more meaningful cognitive engagement.

However, studies including Browning et al. (2024); Hasan et al. (2024) reported negative effects from unstructured or excessive use. In these cases, students often relied on GenAI for quick solutions, leading to superficial engagement and a diminished capacity for critical thinking. Over-reliance was particularly evident among learners with limited prompting skills or foundational knowledge, who tended to accept AI output without scrutiny. Learner ability further mediated these outcomes. Higher-performing students demonstrated strategic use of GenAI for abstraction and planning, while their lower-performing peers focused on syntax-level fixes, showing less critical engagement (Browning et al. (2024); Lai and Lin (2025)). This highlights the importance of cognitive maturity in leveraging AI tools effectively.

Instructor involvement also played a pivotal role. Studies such as Kusam et al. (2024); Sinha et al. (2024) showed that when educators structured and supervised AI use, students exhibited more deliberate and reflective thinking. In contrast, unguided use often led to passive learning behaviors. Finally, the context in which GenAI was applied shaped its cognitive impact. Winarta et al. (2024) and Liu, Yu, et al. (2024) suggest that alignment with task design, assessment rubrics, and classroom culture influences whether GenAI tools enhance or undermine critical engagement.

GenAI tools can support problem-solving and critical thinking when used within structured, scaffolded learning environments that account for learner variability. Without such design, overreliance may promote shallow engagement, particularly among novices. This aligns with Martin et al. (2025), who propose Load Reduction Instruction (LRI) to manage cognitive burden through tailored scaffolding, structured practice, and timely feedback. When guided by such principles, GenAI can act as a cognitive amplifier rather than a shortcut. These findings suggest that the academic performance gains reported in RQ1 are more likely to be sustained when GenAI tools are designed to support learners' problem-solving and critical thinking processes rather than replace them.

The development of problem-solving and critical thinking skills in GenAI-supported environments was consistently associated with learners' metacognitive regulation. Studies indicated that tools scaffolding reflection, monitoring, and iterative reasoning supported deeper problem-solving, whereas unstructured use was linked to weakened regulatory control and surface-level engagement.

5. Discussion

5.1. Overview of findings

This study examines how GenAI-driven adaptivity and hallucination dynamics shape learners' metacognitive regulation processes, including self-monitoring, confidence judgment, and error detection, and how these processes influence problem-solving, learning performance, and engagement in CSE. The discussion also considers how GenAI-supported personalization can promote culturally inclusive and contextually grounded learning experiences.

In CSE, learning involves conceptual understanding, algorithmic reasoning, syntax mastery, and iterative problem-solving through debugging. These domain-specific demands engage multiple cognitive processes, making structured integration of GenAI tools essential for supporting learning outcomes, metacognitive development, and engagement through adaptive personalization, hallucination management, and cultural contextualization.

This review explores key areas including adaptive GenAI tools, user-driven personalization, the effects of AI hallucinations, and the support GenAI platforms provide for culturally diverse learners. The findings highlight the dual nature of GenAI's influence: these tools can enhance cognitive development and learner engagement, but they may also increase cognitive dependency, depending on how they are integrated into learning environments and pedagogical approaches.

5.2. GenAI personalization in CSE (RQ1)

Two dominant personalization strategies emerge from the analysis: adaptive and user-driven personalization. Adaptive GenAI tools yielded consistently positive outcomes when integrated with instructional scaffolding such as formative feedback, difficulty regulation, and task-specific guidance. These findings align with recent empirical evidence by Nathaniel et al. (2025a). Their study showed that structured GenAI integration, supported by scaffolded instructional phases and prompt-engineering strategies, enhances higher-order thinking skills and programming logic while also reducing passive dependence on AI support. Tools such as CodeAid, GPT-based tutors, and CodeTutor were found to improve performance in programming assignments, algorithm design tasks, and debugging exercises, as well as students' conceptual understanding and confidence in computational problem-solving. Adaptive personalization also helps manage the intrinsic cognitive load associated with syntax mastery and iterative code debugging.

In contrast, user-driven personalization showed mixed results, with its effectiveness hinging on a learner's ability to input clear, goal-oriented queries and actively engage with the GenAI system. Of the 35 studies examined to assess the impact of GenAI-based personalized learning tools on academic performance, 19 (54.3%) demonstrated positive outcomes, consistent with the theoretical expectation that adaptive personalization enhances learning by optimizing cognitive load and maintaining engagement.

Studies by Lai and Lin (2025); Kazemitabaar et al. (2024); Lyu et al. (2024) reported that adaptive feedback mechanisms and difficulty regulation significantly improve engagement and cognitive development, particularly in programming education, aligning with Aggarwal (2023). However, user-driven personalization was more effective for experienced learners who could articulate precise prompts (Joshi et al., 2024; Kaleemunnisa et al., 2024; Yang et al., 2024), while novice learners often encountered inconsistent responses, undermining learning. This divergence emphasizes the necessity of structured scaffolding to prevent cognitive overload and over-reliance on GenAI outputs, a concern echoed by Duenas and Ruiz (2024).

5.3. Cognitive effects of hallucinations (RQ2)

Building on these personalization strategies, this study explores the cognitive effects of hallucinations, defined as plausible but incorrect outputs from GenAI systems. These outputs exert both disruptive and developmental influences on learners, categorized into cognitive disruption and metacognitive engagement. On one hand, hallucinations led to misconceptions, over-reliance, and cognitive overload in tasks requiring accuracy, particularly programming exercises and algorithmic reasoning assignments (Husain, 2024; Prather et al., 2024; Ramirez Osorio et al., 2025). On the other hand, when learners were encouraged to question and verify AI-generated outputs, hallucinations triggered metacognitive processes, such as inquiry, reflection, and conceptual reevaluation (Hasan et al., 2024; Kusam et al., 2024; Prather, Reeves, et al., 2023; Suárez-Pizzarello et al., 2024).

These dual impacts are consistent with prior literature on AI-related errors in educational settings. Elsayed (2024) noted that unchecked hallucinations, like fabricated references or false explanations, can undermine trust and learning. However, Jiang et al. (2024) argue that such inaccuracies, when framed pedagogically, can enhance creative problem-solving and foster reflective thinking. Instructor-guided reflection and task design (Dos Santos & Cury, 2023; Yeh et al., 2025) were found to be crucial in transforming hallucinations from sources of confusion into opportunities for deep learning.

Given the dual nature of GenAI hallucinations, either as cognitive distractions or as catalysts for reflection, there remains a pressing need to explore how students engage with hallucinated outputs in authentic classroom settings. While prior studies have acknowledged both the risks and pedagogical opportunities of hallucinations, few have empirically examined their impact on learners' cognitive strategies, problem-solving behavior, and academic performance in real-world environments. Future work will therefore focus on an in-situ investigation to analyze how students interpret, respond to, and learn from hallucinated outputs generated by GenAI tools. The future study will seek to identify not only under what conditions hallucinations hinder learning, but also how structured reflection, prompt literacy, and instructor mediation can transform these challenges into productive learning experiences.

5.4. Cultural and linguistic inclusivity (RQ3)

The findings on inclusivity underscore a developing yet critically important research dimension within GenAI-supported CSE, one that remains underexplored despite its centrality to equitable learning. Although the six studies reviewed provide encouraging evidence of efforts to enhance language access, cultural relevance, and equity, the overall coverage remains uneven. Most studies (Jordan et al., 2024; Prather, Reeves, et al., 2025) addressed inclusivity indirectly through evaluations of language adaptation or contextual quality rather than as a central research objective. Similarly, research on cultural contextualization (del Carpio Gutierrez et al., 2024; Oyelere & Aruleba, 2025) demonstrates that integrating local and culturally relevant examples can significantly improve engagement and perceived relevance, yet such approaches are geographically narrow and methodologically isolated. Moreover, investigations into equity (Maurat et al., 2024; Pang et al., 2024) focus largely on self-reported differences in access, gender, or socioeconomic variables rather than structural or pedagogical interventions that promote inclusion.

Multilingual prompting enabled students to engage more effectively, particularly in native languages such as Arabic, Hausa, Igbo, Spanish, Yoruba, and Tamil. However, significant disparities were observed: GenAI systems provided more accurate and coherent support in high-resource languages, leaving low-resource languages underrepresented. This linguistic imbalance risks reinforcing structural educational inequalities, especially for learners from underserved backgrounds. Cultural contextualization of programming tasks improved learner engagement when aligned authentically with local contexts, superficial adaptations, however, often increased cognitive load and disengagement. Studies by Oyelere and Aruleba (2025) and del Carpio Gutierrez et al. (2024) emphasize the importance of authentic integration rather than superficial inclusion. Similarly, the effectiveness of personalization was shaped by factors such as socioeconomic status (SES), gender, and infrastructure availability. Learners from higher-SES backgrounds used GenAI tools for complex tasks, while those from less privileged backgrounds tended to use them for basic support, amplifying existing educational disparities. These findings echo Guerrero (2024), who argue that pedagogically grounded and culturally aware tools are essential for equitable education.

Future research should adopt cross-cultural, multilingual, and intersectional approaches to systematically examine how GenAI systems can support inclusive participation across diverse learner populations. This includes designing experiments that test linguistic and cultural adaptability at scale, as well as longitudinal studies that assess the long-term equity impacts of GenAI integration in computing education.

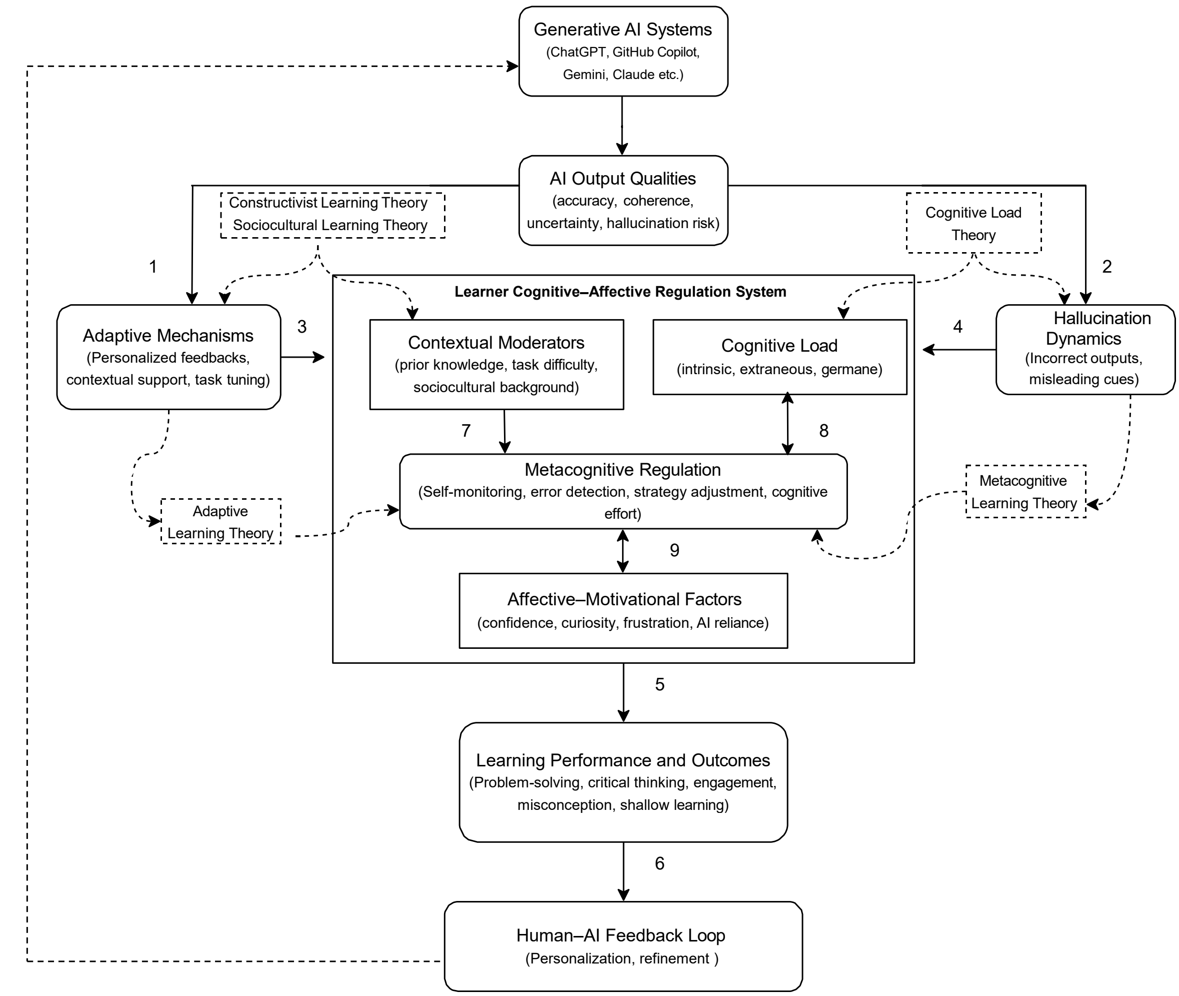
5.5. Problem-solving and critical thinking outcomes (RQ4)

In relation to cognitive development, findings indicate that GenAI tools enhance problem-solving and critical thinking when embedded within well-structured learning environments. Adaptive GenAI tools, such as PyTutor and CodeTailor, provided metacognitive prompts, iterative feedback, and opportunities for reflection. These tools were shown to support the development of higher-order thinking skills by facilitating reasoning and self-regulation, aligning with Borge et al. (2024) and the principles of Load Reduction Instruction (LRI) discussed by Martin et al. (2025). However, unguided or passive use of these tools, as in studies like Browning et al. (2024) and Hasan et al. (2024) encouraged surface-level engagement, where learners focused more on obtaining solutions than understanding underlying concepts. These observations align with foundational adaptive learning frameworks such as those proposed by Cavanagh et al. (2020), which emphasize that meaningful gains occur when real-time feedback, task-specific scaffolding, and active learner involvement are harmonized. GenAI tools must therefore be positioned not as replacements for instruction but as interactive partners in learning, requiring intentional pedagogical design.

5.6. Integrative framework of GenAI-Supported learning

Building upon the synthesized findings across RQ1–RQ4, this section presents the Integrative Framework of GenAI-Supported Learning (Fig. 9). The framework consolidates empirical evidence to illustrate how GenAI-driven adaptivity and hallucination dynamics function as technological mechanisms that shape learners' cognitive and affective regulation processes, such as metacognitive monitoring, interpretation, and strategic adjustment, which in turn influence learning performance and cognitive outcomes in CSE. Constructivist Learning Theory, Sociocultural Learning Theory, Cognitive Load Theory, and Metacognitive Learning Theory collectively provide the theoretical foundation for understanding how learners regulate, interpret, and act upon AI-generated information within these interactions.

At the top of the framework, Generative AI Systems (e.g., ChatGPT, GitHub Copilot, Gemini, Claude) produce a range of AI Output Qualities,



—► Empirically supported association (RQ-linked)
--► Theoretically inferred relationship
↔ Bidirectional or reciprocal association

Arrow #	Source → Target	Supporting RQ(s)
1	AI Output Qualities → Adaptive Mechanisms	RQ1
2	AI Output Qualities → Hallucination Dynamic	RQ2
3	Adaptive Mechanisms → Cognitive–Affective Regulation	RQ1, RQ4
4	Hallucination Dynamics → Cognitive–Affective Regulation	RQ2
5	Cognitive–Affective Regulation → Learning Performance and Outcomes	RQ1, RQ4
6	Learning Performance and Outcomes → Human–AI Feedback Loop	RQ1, RQ2,RQ4
7	Contextual Moderators → Metacognitive Regulation	RQ3
8, 9	Metacognitive Regulation ↔ Cognitive Load / Affective–Motivational Factors	RQ1,RQ4

Fig. 9. Integrative framework of GenAI-Supported learning: Interplay of adaptivity, hallucination dynamics, and metacognitive regulation.

including accuracy, coherence, uncertainty cues, and hallucination risk. These qualities determine how students interpret, trust, or question system-generated content. Constructivist and Sociocultural learning theories support this relationship by explaining how learners interact with AI as both a cognitive scaffold and a sociocultural tool.

From AI Output Qualities, two primary technological pathways emerge: Adaptive Mechanisms and Hallucination Dynamics. Adaptive mechanisms provide personalized feedback, contextualized support, and task tuning, facilitating engagement and comprehension (RQ1). In contrast, hallucination dynamics introduce misleading outputs and incorrect cues (RQ2), which can hinder understanding but may also stimulate verification, reflection, and deeper inquiry when appropriately scaffolded.

These pathways feed into the Learner Cognitive–Affective Regulation System, the central component of the framework. This system contains three interacting layers that shape how learners process and respond to AI output. Contextual Moderators (such as prior knowledge, task difficulty, sociocultural background) influence how learners interpret adaptive support or misleading information. Cognitive Load reflects the mental effort required to process adaptive or misleading AI output. Cognitive Load Theory clarifies how hallucinations can increase extraneous load and disrupt regulatory efficiency. Affective–Motivational

Factors (such as confidence, curiosity, frustration, AI reliance) shape learners' willingness to engage critically with AI responses and determine whether they accept or challenge system outputs.

At the center of this system is Metacognitive Regulation, encompassing learners' monitoring, evaluation, and strategic control of their cognitive effort. Both adaptivity and hallucinations converge here. Adaptive support promotes productive metacognition by reducing unnecessary load and supporting reflection, while hallucinations require learners to evaluate uncertainty, detect inconsistencies, and regulate cognitive load. Metacognitive Learning Theory provides theoretical grounding for this regulatory process.

A reciprocal relationship exists between cognitive load and metacognitive regulation: effective monitoring and strategy adjustment help reduce unnecessary cognitive load, while high load can weaken metacognitive accuracy and slow error detection. A similar two-way influence occurs between affective–motivational factors and metacognitive regulation. Strong regulation supports confidence and reduces frustration, whereas emotions such as uncertainty or overreliance on AI can impair metacognitive processes. These reciprocal interactions highlight the dynamic centrality of metacognitive regulation within GenAI-supported learning.

The effects of these regulatory processes are reflected in Learning Performance and Outcomes, which include problem-solving, critical thinking, engagement, misconception formation, and shallow or deep learning. Findings from RQ1 and RQ4 show that strong performance emerges when adaptivity reduces extraneous load and supports reflective regulation. Conversely, high exposure to hallucinated outputs without scaffolding increases cognitive load and may produce inaccuracies or misconceptions.

Finally, the framework includes a Human–AI Feedback Loop, representing how learner behaviors, errors, and performance patterns contribute to the personalization and refinement of GenAI systems over time. This influence underscores the dynamic relationship between learners and adaptive AI tools. The arrow from the Human–AI Feedback Loop to Generative AI Systems represents how learner behaviors and performance patterns inform system refinements. These refinements, in turn, influence AI Output Qualities, ensuring the feedback loop is fully represented even without a direct arrow to this component. Although learner feedback does not directly update the AI model's parameters, it can influence the AI's subsequent outputs. For example, it may help the system recover from hallucinations or improve response accuracy. This completes the conceptual feedback loop between learners and the AI system. Solid arrows represent empirically supported associations synthesized across multiple studies. Dashed arrows represent theoretically inferred connections, and bidirectional dotted arrows indicate reciprocal associations suggested by the literature. All arrows denote associative, non-causal relationships aligned with specific research questions (RQ1–RQ4), rather than causal effects. This integrative framework demonstrates how adaptivity, hallucination dynamics, cognitive–affective regulation, and learning performance interact to shape the pedagogical role of GenAI in CSE.

5.7. Theoretical foundations underpinning the framework

Within the integrative framework developed in this review, CSE presents distinct learning challenges, including high cognitive load in algorithmic reasoning, difficulties in debugging and error detection, uneven prior knowledge, and increasing reliance on automated tools that may obscure conceptual understanding. The introduction of GenAI intensifies these challenges by introducing adaptive assistance alongside risks such as hallucinated outputs, automation bias, and reduced learner agency. Accordingly, learning theories are not invoked here as abstract pedagogical lenses, but as explanatory tools for understanding how learners interpret, regulate, and respond to AI-generated support and misinformation in CSE contexts. The following theories are therefore mapped directly to specific mechanisms within the framework, clarifying how GenAI-mediated interactions can either support or undermine problem-solving, metacognitive regulation, and learning performance.

This subsection explicates how established learning theories map onto specific components and relationships within the integrative framework, clarifying the theoretical rationale underlying the synthesized associations rather than introducing new empirical claims.

The relationships represented in the framework are supported by five major learning theories: Constructivist Learning Theory, Sociocultural Learning Theory, Cognitive Load Theory, Adaptive Learning Theory, and Metacognitive Learning Theory. These theories explain how learners interpret GenAI output, regulate cognitive and affective processes, and construct knowledge within GenAI-mediated learning environments.

Constructivist Learning Theory posits that learners actively build knowledge by integrating new information with existing mental structures (Guo et al., 2024; Wu, 2023; Zajda, 2021). This perspective explains the learner's active process of building knowledge while interacting with GenAI-generated explanations, examples, and problem-solving cues (Pavlik, 2025). This theory supports the pathway between AI Output Qualities and learners' metacognitive interpretation processes. When GenAI provides coherent explanations, step-by-step reasoning or adaptive support, learners assimilate and accommodate new information through sensemaking. Conversely, hallucinations require reconstructive reasoning, which may evoke deeper inquiry or confusion depending on learners' regulatory capacities. Therefore, constructivism explains the interpretive work learners perform as they evaluate and integrate GenAI-generated content.

Sociocultural Theory emphasizes that learning is mediated by cultural tools, language, and social context (Coppens & Kelley, 2025; Polly et al., 2017; Wu, 2023). Within this framework, GenAI functions as a semiotic mediator, meaning a symbolic or cultural tool such as language, diagrams, or AI-generated explanations that shapes how learners think, interpret information, and engage in problem-solving. Through this mediating role, GenAI influences learners' reasoning strategies, discourse patterns, and problem-solving practices. The contextual moderators in the model, including learners' sociocultural background, prior knowledge, linguistic familiarity, and educational experiences, are central to this theoretical perspective. These factors shape how learners perceive and respond to AI-generated content, including whether they interpret the system as authoritative, collaborative, or fallible. Consequently, Sociocultural Theory helps explain why students from different cultural or educational contexts may interpret identical AI outputs differently, resulting in varied cognitive load profiles, epistemic stances (meaning learners' orientations toward knowledge and knowing, including their beliefs about the reliability of information sources and the criteria they use to evaluate and justify understanding), and metacognitive regulatory responses.

Cognitive Load Theory (Lo et al., 2024; Sweller, 2020; Wu, 2023) aligns with the framework's middle layer, where information processing demands determine how learners absorb and integrate AI-generated content. Accurate, well-structured output reduces extraneous load, allowing learners to allocate more cognitive resources to germane load, which supports schema construction and refinement. In contrast, hallucinated, ambiguous, or misleading responses increase extraneous load and redirect attention away from relevant processes, potentially impairing learning efficiency. In this context, intrinsic load refers to the inherent complexity of the computer science task. Extraneous load reflects unnecessary cognitive effort introduced by poorly structured or incorrect AI output. Germane load represents the productive mental effort devoted to sensemaking, integrating new information, and refining problem-solving strategies. The balance among these three components determines the efficiency of learners' metacognitive monitoring and regulation. This theoretical foundation clarifies how AI output characteristics interact with learners' cognitive capacity to shape their interpretations, decisions, and regulatory actions.

Adaptive Learning approaches (Tarun et al., 2025) provide a framework for understanding how instructional systems tailor feedback, difficulty, and presentation based on learner performance, goals, and behavioral patterns (Corbett & Anderson, 1994). Recent evidence on AI-driven adaptive learning systems reinforces these principles, showing how modern platforms dynamically adjust content, guidance, and learning trajectories to meet individual learner needs (Jawad & Ashiq, 2024; Rincon-Flores et al., 2024; Ruan & Lu, 2025). In the framework, the adaptive pathway reflects GenAI's ability to adjust explanations, problem difficulty, and support in real time. Although adaptivity is conceptually distinct from Sociocultural Learning Theory, its pedagogical value aligns with Vygotsky's notion of maintaining learners within their zone of proximal development, meaning they provide support that keeps tasks appropriately challenging, neither too easy nor too difficult, so that learners can make progress with guidance while continuing to develop new skills (Cai et al., 2025). This alignment illustrates how adaptive responses promote cognitive efficiency, motivation, and sustained engagement. However, adaptivity does not operate in isolation; it interacts with contextual and affective moderators, shaping learners' perceived levels of challenge, frustration, or support. Hence, Adaptive Learning Theory provides the foundation for understanding how personalization influences metacognition and performance within GenAI-mediated learning environments.

Metacognitive theory clarifies how learners monitor comprehension, detect inconsistencies, and regulate strategies during learning (Moshman, 2018; Tankelevitch et al., 2024; Van Velzen, 2016; Xu et al., 2025). In the framework, metacognitive regulation is central to managing the effects of accurate and hallucinated outputs. Learners with strong metacognitive skills question uncertainty cues, verify claims, and adjust strategies when encountering ambiguous or incorrect content. Conversely, learners with limited metacognitive awareness may accept misinformation, misjudge their understanding, or over-rely on AI systems. Metacognitive theory therefore explains the learner-level mechanisms that determine whether GenAI enhances or hinders learning.

These theories illuminate distinct yet interconnected mechanisms within the framework. Sociocultural factors shape learners' initial stance toward GenAI and the interpretive norms they bring to AI-mediated interactions. Cognitive load determines the extent to which learners can process and integrate AI-generated information. Adaptive learning principles account for variation in task personalization and learners' perceived level of challenge. Metacognitive regulation governs the monitoring and control processes that determine whether learners benefit from accurate AI support or compensate effectively for hallucinated output. By embedding these theoretical traditions within the model, the framework provides a robust conceptual foundation for understanding how GenAI tools influence metacognition, cognitive load, motivation, and performance in CSE.

5.8. Evidence robustness and bias risk

The quality appraisal (QA1–QA5) indicates a generally high level of methodological clarity and reporting rigor across the 64 included studies. All studies clearly stated their research aims (QA1), described their methodologies (QA4), and presented their findings in sufficient detail (QA3), supporting confidence in the internal validity of the synthesized results. Additionally, 94% of studies explicitly engaged with core GenAI-related themes (QA2), demonstrating strong conceptual alignment with the scope of this review.

However, several factors temper the robustness of the evidence base. First, QA5 revealed that most uncited studies were recently published, 64% in 2024 and 32% in 2025, suggesting that low citation counts reflect citation immaturity rather than methodological limitations. This recency bias underscores the fast-evolving nature of GenAI research and highlights the need for ongoing monitoring of evidence accumulation.

A potential publication bias was also observed. As presented in Table 8 (Section 4.6.4), 54.3% of studies reported positive academic impacts of GenAI personalization, whereas neutral and negative outcomes each appeared in only 5.7% of cases. Mixed outcomes accounted for 34.3%. This skew toward favorable findings suggests that studies reporting novel or beneficial effects may be more likely to be published, while null or adverse results remain underrepresented in the literature.

Methodological heterogeneity further complicates evidence synthesis. The reviewed studies spanned qualitative classroom investigations, experimental trials, quasi-experimental designs, and survey-based evaluations (Section 4.5). While this diversity provides breadth, it also introduces variability in analytical rigor, operational definitions, and outcome measures, limiting direct comparability across research questions. Evidence strength was notably uneven across RQs: RQ1 (GenAI personalization), RQ2 (hallucination effects), and RQ4 (problem-solving and critical thinking) were supported by a substantial number of methodologically grounded studies. In contrast, RQ3 (inclusivity and cultural dimensions) drew from only a small set of papers, reflecting a structural gap in the field rather than a limitation of this review. This scarcity constrains the generalizability of findings related to cultural and linguistic inclusivity.

To improve evidence robustness, future research should adopt longitudinal or multi-cohort study designs, recruit more diverse learner populations, and implement standardized reporting frameworks to facilitate cross-study compara-bility. Furthermore, greater inclusion of non-Western, multilingual, and low-resource educational contexts is essential for addressing cultural and linguistic biases and ensuring that GenAI applications in CSE are equitable, contextually grounded, and globally relevant.

5.9. Summary of findings

Table 11 synthesizes key findings across all research questions. RQ1 demonstrates that adaptive personalization yields more consistent learning gains than user-driven approaches. RQ2 highlights the dual cognitive role of hallucinations as sources of disruption and metacognitive engagement. RQ3 reveals persistent inequities in linguistic, cultural, and socioeconomic access. RQ4 shows that scaffolded GenAI integration supports problem-solving and critical thinking, whereas unstructured use promotes surface-level engagement.

Beyond thematic patterns, the review also identifies important implications for educational and pedagogical data analytics. GenAI systems produce extensive learner interaction traces, including prompt histories, error patterns, reasoning paths, and feedback cycles, that can support data-driven learner modeling and real-time personalization. These analytics enable instructors to monitor learning behaviors, detect misconceptions early, and interpret metacognitive indicators to inform instructional decisions. However, data-driven personalization also poses risks, such as biased learner models for students from underrepresented linguistic or cultural backgrounds and the potential misinterpretation of opaque AI-generated learning traces. Hence, learning analytics derived from GenAI outputs must be applied transparently, ethically, and in alignment with established pedagogical goals.

Educators should integrate GenAI tools within structured learning activities, providing adaptive scaffolds, guided reflection, and task-specific feedback to support metacognitive regulation, problem-solving, and critical thinking, while mitigating risks of over-reliance or cognitive overload. AI designers and developers should focus on context-aware, adaptive, and culturally relevant systems, supporting multilingual access, signaling uncertainty in outputs, and aligning system behavior with pedagogical objectives. Ethical and inclusive implementation is essential: disparities in language, Socioeconomic status and infrastructure must be addressed, and transparent, culturally authentic, and fair design practices should guide the responsible deployment of GenAI in CSE.

Finally, the synthesis underscores GenAI's innovative role in reshaping educational practice within CSE. Unlike traditional instructional methods grounded in static materials and delayed feedback

Table 11
Summary of findings, quantitative patterns, and implications across RQs (RQ1–RQ4).

RQ	Focus/Theme	Key Findings	Quantitative/Pattern Summary	Implications
RQ1	GenAI personalization	Adaptive (system-driven) personalization improves academic performance via task-specific feedback, adaptive difficulty, and RAG-supported outputs. User-driven personalization shows variable outcomes depending on learner skill, prompt quality, and engagement. Over-reliance can reduce problem-solving autonomy.	35 studies: Positive (19, 54.3%), Mixed (12, 34.3%), Negative (2, 5.7%), Neutral (2, 5.7%). Adaptive strategies linked to stronger gains.	Design GenAI systems with adaptive scaffolds; monitor over-reliance; train students in effective prompt usage; balance system- and user-driven personalization.
RQ2	Cognitive effects of hallucinations	Hallucinations can disrupt cognition (misleading outputs, cognitive overload, over-reliance) but also stimulate metacognitive engagement (reflection, prompt refinement, self-monitoring). Instructor scaffolding reduces risks and enhances reflective thinking.	43 studies: Mixed effects (25), Hindering (9), Stimulating (9). Effects shaped by learner expertise, task complexity, and instructional support.	Incorporate scaffolding, reflection prompts, and metacognitive strategy training to mitigate hallucination risks and promote cognitive engagement.
RQ3	Cultural inclusivity	GenAI can support language diversity (multilingual prompts) and culturally relevant contexts (localized exercises). It also has the potential to promote equity in access and personalization by providing adaptive, learner-centered resources. However, effectiveness is limited by low-resource language coverage, superficial contextualization, socioeconomic status/gender disparities, and infrastructural barriers.	6 studies: Language inclusivity (2), Cultural contextualization (2), Equity in personalization/access (3). One study addressed multiple dimensions. The inclusivity dimensions remain underexplored, with limited empirical coverage across studies.	Develop culturally aware AI tools; prioritize low-resource language support; address access inequalities; adapt content to local contexts.
RQ4	Problem-solving and critical thinking	Structured, scaffolded integration of GenAI tools enhances reflective problem-solving, iterative reasoning, and critical thinking in programming labs, debugging challenges, and algorithm design projects. Unguided or informal use can promote shallow engagement, over-reliance, and uneven cognitive benefits. Instructor involvement mediates outcomes; tool integration depth is critical.	28 studies: Enhanced cognitive outcomes (15), Mixed effects (8), Limited benefits/Over-reliance (5). High-performing students leverage AI for abstraction and planning; low-performing students focus on syntax-level corrections.	Integrate GenAI with guided activities and timely feedback; customize support to learner performance; enhance higher-order reasoning; use GenAI to pro-mote conceptual understanding, not just syntax.

cycles, GenAI introduces dynamic adaptivity, real-time formative support, culturally contextualized examples, and interactive problem-solving guidance. These innovations enable scalable personalization, immediate metacognitive prompting, and data-driven learner modeling previously difficult to achieve at classroom scale. By synthesizing how GenAI's adaptive mechanisms, hallucination dynamics, and inclusivity features jointly influence learning, this review advances a unified, theory-informed perspective that extends beyond existing studies. In doing so, it provides novel insights into how AI-driven tools can transform pedagogical design, learner support, and instructional decision-making in CSE.

6. Conclusion

The integration of GenAI into CSE is transforming pedagogical approaches, redefining learner support systems, and reshaping the role of educators. These technologies offer powerful tools that enable adaptive learning, enhance problem-solving skills, and promote inclusive engagement. This systematic review emphasizes that the educational value of GenAI lies not merely in its generative capabilities, but in how it is ethically and pedagogically embedded within structured learning environments. When thoughtfully implemented and scaffolded, GenAI platforms can promote metacognitive development, support culturally and linguistically diverse learners, and enhance critical thinking. The integrative framework further clarifies that these benefits emerge through the interplay of AI output qualities, adaptive mechanisms, and the learner's cognitive and affective regulation processes, which together shape performance, comprehension, and engagement.

At the same time, this review highlights critical issues. Technological advancement alone does not ensure equitable access, culturally relevant learning, or meaningful engagement. Without deliberate instructional design, GenAI tools risk functioning as mechanisms of automation rather than amplification, reinforcing linguistic, socioeconomic, and pedagogical disparities. The framework shows that hallucination dynamics, cognitive load, and affective responses can interact in ways that undermine learning if left un-scaffolded, particularly when learners misinterpret uncertainty cues or lack strong metacognitive regulation. Addressing these concerns requires a shift from tool adoption to systemic integration that prioritizes learner agency, inclusive design, and continuous, context-aware feedback, while also accounting for AI reliability, transparency, and fairness.

The findings are theoretically grounded in Constructivism, Sociocultural Learning Theory, Cognitive Load Theory, Adaptive Learning Theory, and Metacognitive Regulation models, reflecting the expanded theoretical base of the framework. Constructivism explains how learners actively construct understanding through reflective engagement with GenAI feedback, while Sociocultural theory highlights how contextual moderators shape interpretations of AI output. Cognitive Load Theory clarifies how adaptive mechanisms can reduce extraneous load, whereas hallucinations can elevate it, requiring stronger regulation. Adaptive Learning Theory explains how personalization emerges from interaction patterns, and Metacognitive Learning Theory positions regulation as the central process that determines whether GenAI interactions enhance or hinder cognitive efficiency and learning outcomes. This theoretical integration provides a coherent explanation for the mixed empirical outcomes observed across studies and supports the dynamic interactions represented in the framework's learner regulation system.

Programming, and by extension CSE, relies heavily on syntax mastery, algorithmic reasoning, abstraction, and iterative debugging. Consequently, the integration of GenAI in this domain requires intentional scaffolding to ensure that these tools support meaningful computational learning rather than encourage passive code generation. The framework reinforces this point by illustrating how misinterpretations of AI output or unregulated reliance can disrupt problem-solving and introduce misconceptions when hallucination dynamics are not properly managed.

This review offers a comprehensive synthesis of GenAI's role in CSE; however, several limitations should be acknowledged. Despite adherence to PRISMA and Kitchenham guidelines, the scope was limited to English-language publications indexed in selected databases, potentially excluding relevant studies from non-English-speaking or low-resource contexts. Although most of the included studies were empirical, many were short-term and lacked longitudinal data on sustained learning outcomes. As such, the long-term cognitive, motivational, and ethical impacts of GenAI integration, including how learners' regulation systems adapt over time, remain underexplored.

As GenAI technologies continue to evolve, their role in CSE must be reexamined through empirical inquiry and ethical scrutiny. Future research should investigate long-term impacts, develop inclusive design frameworks, and establish robust pedagogical guidelines to ensure GenAI enhances, not hinders, learning outcomes. In particular, our future work will focus on conducting empirical research to examine how GenAI hallucinations influence students' understanding, reasoning processes, and critical engagement in real-world learning environments. Fundamentally, the transformative potential of GenAI will be determined not merely by what it generates, but by how it engages with learners' cognitive and affective processes to support deeper, equitable, and reflective learning.

CRediT authorship contribution statement

Adedeji Adefisoye Adejumo: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Solomon Sunday Oyelere:** Writing – review & editing, Validation, Supervision, Methodology, Investigation. **Ismaila Temitayo Sanusi:** Writing – review & editing, Validation, Supervision, Methodology, Investigation. **Jarkko Suhonen:** Writing – review & editing, Validation, Supervision, Methodology, Investigation.

Statements on open data and ethics

The data used in this systematic literature review were obtained from publicly accessible scholarly databases and publications. All sources are properly cited in the references section, and detailed citations are provided for each included study. As this review is based solely on secondary data and does not involve human participants, ethical approval and informed consent were not required.

Funding

No funding was received for this work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Abdulla, S., Ismail, S., Fawzy, Y., & Elhag, A. (2024). Using ChatGPT in teaching computer programming and studying its impact on students performance. *Electronic Journal of e-Learning, 22*, 66–81.

Agbo, F. J., Olivia, C., Oguibe, G., Sanusi, I. T., & Sani, G. (2025). Computing education using generative artificial intelligence tools: A systematic literature review. *Computers and Education Open, 9*, Article 100266. https://doi.org/10.1016/j.caeo.2025.100266

Aggarwal, D. (2023). Integration of innovative technological developments and AI with education for an adaptive learning pedagogy. *China Petroleum Processing and Petrochemical Technology, 23*, 709–714.

Ahmad, Z., Kaiser, W., & Rahim, S. (2023). Hallucinations in ChatGPT: An unreliable tool for learning. *Rupkatha Journal on Interdisciplinary Studies in Humanities, 15*, 12.

Amaya, E. J. C., Restrepo-Calle, F., & Ramírez-Echeverry, J. J. (2023). Discovering insights in learning analytics through a mixed-methods framework: Application to computer programming education. *Journal of Information Technology Education: Research, 22*, 339–372.

Anis, M. (2023). Leveraging artificial intelligence for inclusive English language teaching: Strategies and implications for learner diversity. *Journal of Multidisciplinary Educational Research, 12*, 54–70.

Apiola, M., López-Pernas, S., & Saqr, M. (Eds.). (2023). *Past, present and future of computing education research: A global perspective*. Cham, Switzerland: Springer. https://doi.org/10.1007/978-3-031-25336-2.

Asbai, A. (2024). *Mitigating hallucination in large language model code generation for higher education: An evaluation of retrieval augmented generation. Msc thesis KTH royal institute of technology, school of electrical engineering and computer science*.

Axelsson, A., Wallgren, D. T., Verma, U., Cajander, Å., Daniels, M., Eckerdal, A., & McDermott, R. (2024). From assistance to misconduct: Unpacking the complex role of generative AI in student learning. In *2024 IEEE frontiers in education conference (FIE 2024)* (pp. 13–16). Washington, DC, USA: IEEE. https://doi.org/10.1109/FIE61694.2024.10893133. article number: 10893133.

Baláž, N., Porubän, J., Horváth, M., & Kormaník, T. (2024). Using ChatGPT during implementation of programs in education. In *5th international computer programming education conference (ICPEC 2024)*. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 18–1.

Boguslawski, S., Deer, R., & Dawson, M. G. (2025). Programming education and learner motivation in the age of generative AI: Student and educator perspectives. *Information and Learning Sciences, 126*, 91–109.

Borge, M., Smith, B. K., & Aldemir, T. (2024). Using generative AI as a simulation to support higher-order thinking. *International Journal of Computer-Supported Collaborative Learning, 19*, 479–532.

Brender, J., El-Hamamsy, L., Mondada, F., & Bumbacher, E. (2024). Who's helping who? When students use ChatGPT to engage in practice lab sessions. In *Artificial intelligence in education: 25th international conference, AIED 2024, recife, Brazil, July 8–12, 2024, proceedings, part I* (pp. 235–249). Cham, Switzerland: Springer. https://doi.org/10.1007/978-3-031-64302-6_17. volume 14557 of *Lecture Notes in Artificial Intelligence*.

Browning, J. W., Bustard, J., Anderson, N., & Galway, L. (2024). A data science course utilizing GenAI. In *2024 IEEE frontiers in education conference (FIE)* (pp. 1–7). IEEE.

Cai, W., & Gao, M. (2025). Beyond hallucination: Generative AI as a catalyst for human creativity and cognitive evolution. *IEICE Transactions on Emerging Topics in Artificial Intelligence, 2*, 36–42.

Cai, L., Msafiri, M. M., & Kangwa, D. (2025). Exploring the impact of integrating AI tools in higher education using the zone of proximal development. *Education and Information Technologies, 30*, 7191–7264.

Cavanagh, T., Chen, B., Lahcen, R. A. M., & Paradiso, J. R. (2020). Constructing a design framework and pedagogical approach for adaptive learning in higher education: A practitioner's perspective. *International Review of Research in Open and Distance Learning, 21*, 173–197.

Chang, C.-K., & Chien, L.-C.-T. (2024). Enhancing academic performance with generative AI-Based quiz platform. In *2024 IEEE international conference on advanced learning technologies (ICALT)* (pp. 193–195). Nicosia: North Cyprus. https://doi.org/10.1109/ICALT61570.2024.00062. Cyprus: IEEE.

Choudhury, S., Saha, P., Ray, S., Das, S., & Das, D. (2024). Alphaintellect at semeval-2024 task 6: Detection of hallucinations in generated text. In *Proceedings of the 18th international workshop on semantic evaluation (SemEval-2024)* (pp. 952–958).

Coppens, A. D., & Kelley, R. (2025). Building from sociocultural learning theory to culturally responsive assessment. In C. M. Evans, & C. S. Taylor (Eds.), *Culturally responsive assessment in classrooms and large-scale contexts: Theory, research, and practice* (pp. 17–33). London, UK: Routledge. https://doi.org/10.4324/9781003392217-3.

Corbett, A. T., & Anderson, J. R. (1994). Knowledge tracing: Modeling the acquisition of procedural knowledge. *User Modeling and User-Adapted Interaction, 4*, 253–278.

Crandall, A. S., Sprint, G., & Fischer, B. (2023). Generative pre-trained transformer (GPT) models as a code review feedback tool in computer science programs. *Journal of Computing Sciences in Colleges, 39*, 38–47.

Cubillos, C., Mellado, R., Cabrera-Paniagua, D., & Urra, E. (2025). Generative artificial intelligence in computer programming: Does it enhance learning, motivation, and the learning environment? *IEEE Access, 13*, 40438–40455.

de Kereki, I. F., & Garrido, I. (2024). Solving computer science 2 tasks: Students' reflections on the use of ChatGPT. In *2024 IEEE global engineering education conference (EDUCON)* (pp. 1–5). IEEE.

Dechavez, C. F. J. (2024). Contextualizing learning: A multi-variable analysis of student characteristics, educational settings, and academic success. *International Journal of Research and Scientific Innovation, 11*, 228–243.

del Carpio Gutierrez, A., Denny, P., & Luxton-Reilly, A. (2024). Evaluating automatically generated contextualised programming exercises. *Proceedings of the 55th ACM Technical Symposium on Computer Science Education, 1*, 289–295. New York, NY, USA: ACM.

Denny, P., MacNeil, S., Savelka, J., Porter, L., & Luxton-Reilly, A. (2024). Desirable characteristics for AI teaching assistants in programming education. *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V., 1*, 408–414. Association for Computing Machinery.

Deriba, F., Sanusi, I. T., Campbell, O. O., & Oyelere, S. S. (2024). Computer programming education in the age of generative AI: Insights from empirical research. *SSRN Electronic Journal*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4891302.

Deshpande, S., & Szefer, J. (2023). Analyzing ChatGPT's aptitude in an introductory computer engineering course. In *2023 congress in computer science, computer engineering, & applied computing (CSCE)* (pp. 1034–1038). IEEE.

Dos Santos, O. L., & Cury, D. (2023). Challenging the confirmation bias: Using ChatGPT as a virtual peer for peer instruction in computer programming education. In *2023 IEEE frontiers in education conference (FIE)* (pp. 1–7). IEEE.

Drosos, I., Sarkar, A., Xu, X. T., & Toronto, N. (2025). "it makes you think": Provocations help restore critical thinking to AI-assisted knowledge work. arXiv preprint arXiv: 2501.17247 https://arxiv.org/abs/2501.17247.

Duenas, T., & Ruiz, D. (2024). The risks of human overreliance on large language models for critical thinking. *Preprint.* https://doi.org/10.13140/RG.2.2.26002.06082

El-Sayed, O. (2023). Ethical and culturally informed approaches to AI in education, healthcare, and business. *Emerging Trends in Machine Intelligence and Big Data, 15*, 20–26.

Elsayed, H. (2024). The impact of hallucinated information in large language models on student learning outcomes: A critical examination of misinformation risks in AI-assisted education. *Northern Reviews on Algorithmic Research, Theoretical Computation, and Complexity, 9*, 11–23.

Falagas, M. E., Pitsouni, E. I., Malietzis, G. A., & Pappas, G. (2008). Comparison of PubMed, scopus, web of science, and google scholar: Strengths and weaknesses. *The FASEB Journal, 22*, 338–342.

Fariani, R. I., Junus, K., & Santoso, H. B. (2023). A systematic literature review on personalised learning in the higher education context. *Technology, Knowledge and Learning, 28*, 449–476.

Feldman, P., Foulds, J. R., & Pan, S. (2023). Trapping LLM hallucinations using tagged context prompts. arXiv preprint arXiv:2306.06085 https://arxiv.org/abs/2306.06085.

Feng, T. H., Luxton-Reilly, A., Wünsche, B. C., & Denny, P. (2025). From automation to cognition: Redefining the roles of educators and generative AI in computing education. In *Proceedings of the 27th australasian computing education conference* (pp. 164–171).

Fenu, G., Galici, R., Marras, M., & Reforgiato, D. (2024). Exploring student interactions with AI in programming training. In *Adjunct proceedings of the 32nd ACM conference on user modeling, adaptation and personalization (UMAP adjunct '24)* (pp. 555–560). https://doi.org/10.1145/3631700.3665227. Cagliari, Italy: ACM.

Finnie-Ansley, J., Denny, P., Becker, B. A., Luxton-Reilly, A., & Prather, J. (2022). The robots are coming: Exploring the implications of openai codex on introductory programming. In *Proceedings of the 24th australasian computing education conference* (pp. 10–19).

Gabbay, H., & Cohen, A. (2024). Combining LLM-generated and test-based feedback in a mooc for programming. In *Proceedings of the eleventh ACM conference on learning@ scale* (pp. 177–187).

Gabriella, A., Gui, A., & Chanda, R. C. (2024). The use of chatbot and its impact on academic achievement. In *2024 IEEE symposium on industrial electronics & applications (ISIEA)* (pp. 1–6). IEEE.

Goode, J., Skorodinsky, M., Hubbard, J., & Hook, J. (2020). Computer science for equity: Teacher education, agency, and statewide reform. *Frontiers in Education*, 162. Frontiers Media SA volume 4.

Gorson Benario, J., Marroquin, J., Chan, M. M., Holmes, E. D., & Mejia, D. (2025). Unlocking potential with generative AI instruction: Investigating mid-level software development student perceptions, behavior, and adoption. *Proceedings of the 56th ACM Technical Symposium on Computer Science Education, 1*, 395–401.

Guerrero, M. B. (2024). Pedagogical transformation in the age of AI: Advancing toward a more inclusive and sustainable education. *NeuroPedagogy Research Journal, 1*, 43–60.

Guettala, M., Bourekkache, S., Kazar, O., & Harous, S. (2024). Generative artificial intelligence in education: Advancing adaptive and personalized learning. *Acta Informatica Pragensia, 13*, 460–489.

Guo, H., Yi, W., & Liu, K. (2024). Enhancing constructivist learning: The role of generative AI in personalized learning experiences. In *Proceedings of the 26th international conference on enterprise information systems* (pp. 767–770). SCITEPRESS-Science and Technology Publications.

Hamzah, H., Hamzah, M. I., & Zulkifli, H. (2022). Systematic literature review on the elements of metacognition-based higher order thinking skills (hots) teaching and learning modules. *Sustainability, 14*, 813.

Harris, J., Spina, N., Ehrich, L., & Smeed, J. (2013). Literature review: Student-centred schools make the difference. Technical report Australian institute for teaching and school leadership. https://eprints.qut.edu.au/69161/.

Hasan, A., Simion, B., & Pop, F. (2024). Analyzing the uses and perceptions of computer science students towards generative AI tools. In *, Vol. 19. Proceedings of the international conference on virtual learning* (pp. 363–372). https://doi.org/10.58503/icvl-v19y202430

Hayat, A., Akil, S. W., Martinez, H., Khan, B., & Hasan, M. R. (2024). Board 268: Enhancing zero-shot learning of large language models for early forecasting of STEM performance. In *2024 ASEE annual conference & exposition (p. Board 268)*. American Society for Engineering Education.

Hazzan, O., & Erez, Y. (2025). Rethinking computer science education in the age of genai. *ACM Transactions on Computing Education, 25*, 1–9.

Holstein, K., Aleven, V., & Rummel, N. (2020). A conceptual framework for human–AI hybrid adaptivity in education. In *Artificial intelligence in education: 21st international conference, AIED 2020, Ifrane, Morocco, July 6–10, 2020, proceedings, part I* (Vol. 21, pp. 240–254). Springer.

Hoq, M., Rao, A., Jaishankar, R., Piryani, K., Janapati, N., Vandenberg, J., Mott, B., Norouzi, N., Lester, J., & Akram, B. (2025). *Automated identification of logical errors in programs: Advancing scalable analysis of student misconceptions*. arXiv preprint arXiv: 2505.10913.

Hou, X., Wu, Z., Wang, X., & Ericson, B. J. (2024). Codetailor: LLM-Powered personalized parsons puzzles for engaging support while learning programming. In *Proceedings of the eleventh ACM conference on learning@ scale* (pp. 51–62).

Hsu, H.-P. (2025). From programming to prompting: Developing computational thinking through large language model-based generative artificial intelligence. *TechTrends, 69*, 485–506.

Hu, M., Assadi, T., & Mahroeian, H. (2023). Explicitly introducing ChatGPT into first-year programming practice: Challenges and impact. In *2023 IEEE international conference on teaching, assessment and learning for engineering (TALE)* (pp. 1–6). IEEE.

Huang, A. Y., Lin, C.-Y., Su, S.-Y., & Yang, S. J. (2025). The impact of genai-enabled coding hints on students' programming performance and cognitive load in an srl-based python course. *British Journal of Educational Technology, 56*, 1942–1972.

Husain, A. (2024). Potentials of ChatGPT in computer programming: Insights from programming instructors. *Journal of Information Technology Education: Research, 23*, 2.

Iniesto, F., & Bossu, C. (2023). Equity, diversity, and inclusion in open education: A systematic literature review. *Distance Education, 44*, 694–711.

Jawad, M. M., & Ashiq, S. (2024). Harnessing the power of artificial intelligence for adaptive learning systems: A systematic review. *International Journal of Education, 14*, 12–22.

Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y. J., Madotto, A., & Fung, P. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys, 55*, 1–38.

Jiang, X., Tian, Y., Hua, F., Xu, C., Wang, Y., & Guo, J. (2024). A survey on large language model hallucination via a creativity perspective. arXiv preprint arXiv: 2402.06647 https://arxiv.org/abs/2402.06647.

Jordan, M., Ly, K., & Soosai Raj, A. G. (2024). Need a programming exercise generated in your native language? ChatGPT's got your back: Automatic generation of non-english programming exercises using openai GPT-3.5. *Proceedings of the 55th ACM Technical Symposium on Computer Science Education, 1*, 618–624. New York, NY, USA: ACM.

Joshi, I., Budhiraja, R., Dev, H., Kadia, J., Ataullah, M. O., Mitra, S., Akolekar, H. D., & Kumar, D. (2024). ChatGPT in the classroom: An analysis of its strengths and weaknesses for solving undergraduate computer science questions. *Proceedings of the 55th ACM Technical Symposium on Computer Science Education, 1*, 625–631.

Jošt, G., Taneski, V., & Karakatič, S. (2024). The impact of large language models on programming education and student learning outcomes. *Applied Sciences, 14*, 4115. https://doi.org/10.3390/app14104115

Kaleemunnisa, F., Scharff, C., Bathula, K., & Zhumakova, B. (2024). ChatGPT in the classroom: Experimentation in a python class for non-computing majors. In *2024 IEEE integrated STEM education conference (ISEC)* (pp. 1–7). IEEE.

Kamalov, F., Santandreu Calonge, D., & Gurrib, I. (2023). New era of artificial intelligence in education: Towards a sustainable multifaceted revolution. *Sustainability, 15*, Article 12451.

Kazemitabaar, M., Ye, R., Wang, X., Henley, A. Z., Denny, P., Craig, M., & Grossman, T. (2024). Codeaid: Evaluating a classroom deployment of an llm-based programming assistant that balances student and educator needs. In *Proceedings of the 2024 CHI conference on human factors in computing systems* (pp. 1–20). New York, NY, USA: ACM. https://doi.org/10.1145/3613904.3642773.

Khan, M. M. R. R., Habib, S. B., Tasnim, S. T., & Islam, M. A. (2023). Educational AI and ethical growth: Exploring the effects of ChatGPT on student learning strategies, critical thinking, and academic ethics from a bangladeshi academic perspective. In *2023 26th international conference on computer and information technology (ICCIT)* (pp. 1–6). IEEE Cox's Bazar, Bangladesh: IEEE. https://doi.org/10.1109/ICCIT60459.2023.10441564.

Kim, H., & Lee, S. W. (2024). Investigating the effects of generative-AI responses on user experience after AI hallucination. In *Proceedings of the MBP 2024 Tokyo international conference on management & business practices* (pp. 92–101).

Kitchenham, B., & Charters, S. (2007). *Guidelines for performing systematic literature reviews in software engineering. technical report EBSE-2007-01 EBSE technical report*. Keele University and University of Durham.

Kohen-Vacs, D., Usher, M., & Jansen, M. (2025). Integrating generative AI into programming education: Student perceptions and the challenge of correcting AI errors. *International Journal of Artificial Intelligence in Education*, 1–19.

Kouo, J. L. (2022). Don't assess a fish by its ability to climb a tree: Considerations and strategies to ensure equitable formative assessment practices for all learners. In *Research anthology on physical and intellectual disabilities in an inclusive society* (p. 17). Hershey, PA, USA: IGI Global. https://doi.org/10.4018/978-1-6684-3542-7.ch048.

Kumar, A. N., Raj, R. K., Aly, S. G., Anderson, M. D., Becker, B. A., Blumenthal, R. L., Eaton, E., Epstein, S. L., Goldweber, M., Jalote, P., Lea, D., Oudshoorn, M., Pias, M., Reiser, S., Servin, C., Simha, R., Winters, T., & Xiang, Q. (2024). *Computer science curricula 2023. technical report the joint task force on computer science curricula*. New York, NY, United States. https://doi.org/10.1145/3664191.

Kurdi, G., Leo, J., Parsia, B., Sattler, U., & Al-Emari, S. (2020). A systematic review of automatic question generation for educational purposes. *International Journal of Artificial Intelligence in Education, 30*, 121–204.

Kusam, V. A., Shrestha, S., Kattan, K., Maxim, B., & Song, Z. (2024). A pbl-based mini course module for teaching computer science students to utilize generative AI for enhanced learning. In *2024 IEEE frontiers in education conference (FIE)* (pp. 1–9). Washington, DC, USA: IEEE. https://doi.org/10.1109/FIE61694.2024.10893118.

Kwak, M., Jenkins, J., & Kim, J. (2023). Adaptive programming language learning system based on generative AI. *Issues in Information Systems, 24*.

Lai, C.-H., & Lin, C.-Y. (2025). Analysis of learning behaviors and outcomes for students with different knowledge levels: A case study of intelligent tutoring system for coding and learning (its-cal). *Applied Sciences, 15*, 1922. https://doi.org/10.3390/app15041922

Lan, M., & Zhou, X. (2025). A qualitative systematic review on AI empowered self-regulated learning in higher education. *npj Science of Learning, 10*, 21.

Lee, J., Hung, J.-T., Soylu, M. Y., Popescu, D., Cui, C. Z., Grigoryan, G., Joyner, D. A., & Harmon, S. W. (2025). *Socratic mind: Impact of a novel genai-powered assessment tool on student learning and higher-order thinking*. arXiv preprint arXiv:2509.16262.

Leinonen, J., Hellas, A., Sarsa, S., Reeves, B., Denny, P., Prather, J., & Becker, B. A. (2023). Using large language models to enhance programming error messages. *Proceedings of the 54th ACM Technical Symposium on Computer Science Education, 1*, 563–569.

Liang, X., Song, S., Zheng, Z., Wang, H., Yu, Q., Li, X., Li, R.-H., Wang, Y., Wang, Z., Xiong, F., et al. (2024). Internal consistency and self-feedback in large language models: A survey. arXiv preprint arXiv:2407.14507 https://arxiv.org/abs/2407.14507.

Liao, J., Zhong, L., Zhe, L., Xu, H., Liu, M., & Xie, T. (2024). Scaffolding computational thinking with ChatGPT. *IEEE Transactions on Learning Technologies, 17*, 1628–1642.

Liffiton, M., Sheese, B. E., Savelka, J., & Denny, P. (2023). Codehelp: Using large language models with guardrails for scalable support in programming classes. In *Proceedings of the 23rd koli calling international conference on computing education research* (pp. 1–11).

Liu, Y. (2024). Integrating conversational large language models into student learning: A case study of ChatGPT in software engineering education. In *2024 IEEE frontiers in education conference (FIE)* (pp. 1–8). IEEE.

Liu, S., Guo, X., Hu, X., & Zhao, X. (2024). Advancing generative intelligent tutoring systems with GPT-4: Design, evaluation, and a modular framework for future learning platforms. *Electronics, 13*, 4876.

Liu, F., Liu, Y., Shi, L., Huang, H., Wang, R., Yang, Z., Zhang, L., Li, Z., & Ma, Y. (2024). Exploring and evaluating hallucinations in LLM-powered code generation. arXiv preprint arXiv:2404.00971 https://arxiv.org/abs/2404.00971.

Liu, X., Xiao, Y., & Li, D. (2025). Assessing strategic use of artificial intelligence in self-regulated learning: Instrument development and evidence from Chinese university students. *International Journal of Educational Technology in Higher Education, 22*, 69.

Liu, S., Yu, Z., Huang, F., Bulbulia, Y., Bergen, A., & Liut, M. (2024). Can small language models with retrieval-augmented generation replace large language models when learning computer science? *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V., 1*, 388–393. Association for Computing Machinery.

Lo, L., Li, J.-M., & Wu, C.-H. (2024). Journeying alongside generative AI: Integrating perspectives from vicarious learning and cognitive load theory. In *Proceedings of the australasian conference on information systems (ACIS)*. Canberra, Australia.

Luxton-Reilly, A., Simon, A. I., Becker, B. A., Giannakos, M., Kumar, A. N., Ott, L., Paterson, J., Scott, M. J., Sheard, J., et al. (2018). Introductory programming: A systematic literature review. In *Proceedings companion of the 23rd annual ACM conference on innovation and technology in computer science education* (pp. 55–106).

Lyu, W., Wang, Y., Chung, T., Sun, Y., & Zhang, Y. (2024). Evaluating the effectiveness of LLMs in introductory computer science education: A semester-long field study. In *Proceedings of the eleventh ACM conference on learning@ scale* (pp. 63–74).

Ma, B., Li, H., Li, G., Chen, L., Tang, C., Xie, Y., Gu, C., Shimada, A., & Konomi, S. (2025). *Scaffolding metacognition in programming education: Understanding student-ai interactions and design implications*. arXiv preprint arXiv:2511.04144.

Mahaini, D. (2024). *Generative AI in computer science education: A study on academic performance. B.S. thesis University of Twente*.

Mahboob, K., Asif, R., & Umme, L. (2024). Leveraging generative AI for cross-cultural knowledge exchange in higher education. In *Facilitating global collaboration and knowledge sharing in higher education with generative AI* (pp. 186–206). https://doi.org/10.4018/979-8-3693-0487-7.ch008. IGI Global.

Maity, S., & Deroy, A. (2024). Generative AI and its impact on personalized intelligent tutoring systems. arXiv preprint arXiv:2410.10650 https://arxiv.org/abs/2410.10650.

Mak, J., Nakatumba-Nabende, J., Clear, T., Clear, A., Albluwi, I., Andrei, O., Angeli, L., MacNeil, S., Oyelere, S. S., Rattigan, M. H., Sheard, J., & Zhu, T. (2025). Navigating the ethical and societal impacts of generative AI in higher computing education. *arXiv preprint arXiv:2511.15768*. URL: https://arxiv.org/abs/2511.15768.

Maleki, N., Padmanabhan, B., & Dutta, K. (2024). AI hallucinations: A misnomer worth clarifying. In *2024 IEEE conference on artificial intelligence (CAI)* (pp. 133–138). IEEE.

Martin, A. J., Collie, R. J., Kennett, R., Liu, D., Ginns, P., Sudimantara, L. B., Dewi, E. W., & Rüschenpöhler, L. G. (2025). Integrating generative AI and load reduction instruction to individualize and optimize students' learning. *Learning and Individual Differences, 121*, Article 102723.

Martín-Martín, A., Thelwall, M., Orduna-Malea, E., & Delgado López-Cózar, E. (2021). Google scholar, microsoft academic, scopus, dimensions, web of science, and OpenCitations' COCI: A multidisciplinary comparison of coverage via citations. *Scientometrics, 126*, 871–906.

Matelsky, J. K., Parodi, F., Liu, T., Lange, R. D., & Kording, K. P. (2023). A large language model-assisted education tool to provide feedback on open-ended responses. *arXiv preprint arXiv:2308.02439*. https://arxiv.org/abs/2308.02439.

Matobobo, C., Ncube, P. D. N., Ngesimani, N. L., Dzvapatsva, G. P., & Chinhamo, E. (2025). Enhancing computational thinking and problemsolving in programming education through generative AI: A scoped review. In *2025 IEEE global engineering education conference (EDUCON)* (pp. 1–9). IEEE.

Maurat, J. I. C., Isip, E. V., & Lumabas, A. G. R. (2024). A comparative study of gender differences in the utilization and effectiveness of AI-assisted learning tools in programming among university students. In *Proceedings of the 2024 international conference on artificial intelligence and teacher education* (pp. 30–34).

Moshman, D. (2018). Metacognitive theories revisited. *Educational Psychology Review, 30*, 599–606.

Mukherjee, A., & Chang, H. (2023). The creative frontier of generative AI: Managing the novelty-usefulness tradeoff. *arXiv preprint arXiv:2306.03601*. https://arxiv.org/abs/2306.03601.

Nagori, V., Mahmud, S., Zhou, S., & Iyer, L. (2025). Measuring cognitive load among students during gen AI-enabled data analytics problem-solving. In *Proceedings of the 2025 Pre-ICIS SIGDSA symposium* (pp. 1–7). Association for Information Systems. Open access.

Namoun, A., & Alshanqiti, A. (2020). Predicting student performance using data mining and learning analytics techniques: A systematic literature review. *Applied Sciences, 11*, 237.

Nathaniel, J., Oyelere, S. S., Suhonen, J., & Tedre, M. (2025a). Investigating the impact of generative AI integration on the sustenance of higher order thinking skills and understanding of programming logic in programming education. *Computers and Education: Artificial Intelligence*, Article 100460.

Nathaniel, J., Oyelere, S. S., Suhonen, J., & Tedre, M. (2025b). Literature review on the integration of generative AI in programming education. *International Journal of Artificial Intelligence in Education*, 1–32.

National Academies of Sciences, Engineering, and Medicine. (2021). *Cultivating interest and competencies in computing: Authentic experiences and design factors*. National Academies Press. https://doi.org/10.17226/25912

Normadhi, N. B. A., Shuib, L., Nasir, H. N. M., Bimba, A., Idris, N., & Balakrishnan, V. (2019). Identification of personal traits in adaptive learning environment: Systematic literature review. *Computers & Education, 130*, 168–190.

Nyaaba, M., & Zhai, X. (2025). *Developing custom gpts for education: Bridging cultural and contextual divide in generative AI. Preprint].ResearchGate*. https://doi.org/10.13140/RG, 2.

Nyaaba, M., Zhai, X., & Faison, M. Z. (2024). Generative AI for culturally responsive science assessment: A conceptual framework. *Education Sciences, 14*, 1325.

Ogunleye, B., Zakariyyah, K. I., Ajao, O., Olayinka, O., & Sharma, H. (2024). A systematic review of generative AI for teaching and learning practice. *Education Sciences, 14*, 636.

Okafor, D. O., Sanusi, I. T., & Oyelere, S. S. (2025). Afriml: An interactive and culturally-infused tool for teaching machine learning in schools. *Proceedings of the 30th ACM Conference on Innovation and Technology in Computer Science Education V., 1*, 23–29.

Osadchyi, V. V., Krasheninnik, I., Spirin, O., Koniukhov, S., & Diuzhykova, T. (2020). Personalized and adaptive ict-enhanced learning: A brief review of research from 2010 to 2019. In *Proceedings of the 16th international conference on ICT in education, research and industrial applications. Integration, harmonization and knowledge transfer 2732* (pp. 559–571). CEUR Workshop Proceedings.

Ouh, E. L., Gan, B. K. S., Jin Shim, K., & Wlodkowski, S. (2023). ChatGPT, can you generate solutions for my coding exercises? An evaluation on its effectiveness in an undergraduate java programming course. *Proceedings of the 2023 Conference on Innovation and Technology in Computer Science Education V., 1*, 54–60.

Oviedo-Trespalacios, O., Peden, A. E., Cole-Hunter, T., Costantini, A., Haghani, M., Rod, J., Kelly, S., Torkamaan, H., Tariq, A., Newton, J. D. A., et al. (2023). The risks of using ChatGPT to obtain common safety-related information and advice. *Safety Science, 167*, Article 106244.

Oyelere, S. S., & Aruleba, K. (2025). A comparative study of student perceptions on generative AI in programming education across Sub-Saharan Africa. *Computers and Education Open, 8*, Article 100245. https://doi.org/10.1016/j.caeo.2024.100245

Oyelere, S. S., Sanusi, I. T., Agbo, F. J., Oyelere, A. S., Omidiora, J. O., Adewumi, A. E., & Ogbebor, C. (2022). Artificial intelligence in African schools: Towards a contextualized approach. In *2022 IEEE global engineering education conference (EDUCON)* (pp. 1577–1582). IEEE.

Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., … Moher, D. (2021). The prisma 2020 statement: An updated guideline for reporting systematic reviews. *BMJ, 372*, Article n71. https://doi.org/10.1136/bmj.n71

Pang, A., Padiyath, A., Viramontes Vargas, D., & Ericson, B. J. (2024). Examining the relationship between socioeconomic status and beliefs about large language models in an undergraduate programming course. *Proceedings of the 2024 on ACM Virtual Global Computing Education Conference, 1*, 172–178.

Pankiewicz, M., & Baker, R. S. (2023). Large language models (GPT) for automating feedback on programming assignments. *arXiv preprint arXiv:2307.00150*. https://arxiv.org/abs/2307.00150.

Park, A., & Kim, T. (2025). Code suggestions and explanations in programming learning: Use of ChatGPT and performance. *International Journal of Management in Education, 23*, Article 101119.

Patel, H. (2024). Bane and boon of hallucinations in context of generative AI. https://doi.org/10.36227/techrxiv.171198062.20183635/v1. techRxiv Preprint.

Pavlik, J. V. (2025). Considering the pedagogical benefits of generative artificial intelligence in higher education: Applying constructivist learning theory. In *Generative AI in higher education* (pp. 46–58). Edward Elgar Publishing.

Pesovski, I., Santos, R., Henriques, R., & Trajkovik, V. (2024). Generative AI for customizable learning experiences. *Sustainability, 16*, 3034.

Polly, D., Allman, B., Casto, A., & Norwood, J. (2017). Sociocultural perspectives of learning. In R. E. West (Ed.), *Foundations of learning and instructional design technology*. Orem, UT, USA: EdTech Books https://edtechbooks.org/lidtfoundations/sociocultural_perspectives_of_learningavailableonline.

Prather, J., Denny, P., Leinonen, J., Becker, B. A., Albluwi, I., Craig, M., Keuning, H., Kiesler, N., Kohn, T., Luxton-Reilly, A., MacNeil, S., Petersen, A., Pettit, R., Reeves, B. N., & Savelka, J. (2023). The robots are here: Navigating the generative AI revolution in computing education. In *ITiCSE-WGR 2023 workshop proceedings* (pp. 108–159). New York, NY, USA: ACM.

Prather, J., Leinonen, J., Kiesler, N., Gorson Benario, J., Lau, S., MacNeil, S., Norouzi, N., Opel, S., Pettit, V., Porter, L., et al. (2025). *Beyond the hype: A comprehensive review of current trends in generative AI research, teaching practices, and tools. 2024 working group reports on innovation and technology in computer science education*, 300–338.

Prather, J., Reeves, B. N., Denny, P., Becker, B. A., Leinonen, J., Luxton-Reilly, A., Powell, G., Finnie-Ansley, J., & Santos, E. A. (2023). "it's weird that it knows what i

want": Usability and interactions with copilot for novice programmers. *ACM Transactions on Computer-Human Interaction, 31*, 1–31.
Prather, J., Reeves, B. N., Denny, P., Leinonen, J., MacNeil, S., Luxton-Reilly, A., Orvalho, J., Alipour, A., Alfageeh, A., Amarouche, T., et al. (2025). Breaking the programming language barrier: Multilingual prompting to empower non-native english learners. In *Proceedings of the 27th australasian computing education conference* (pp. 74–84). https://doi.org/10.1145/3716640.3716649. New York, NY, USA: ACM.
Prather, J., Reeves, B. N., Leinonen, J., MacNeil, S., Randrianasolo, A. S., Becker, B. A., Kimmel, B., Wright, J., & Briggs, B. (2024). The widening gap: The benefits and harms of generative AI for novice programmers. *Proceedings of the 2024 ACM Conference on International Computing Education Research-, 1*, 469–486.
Qi, J., Zhao, H., Xiang, R., Zhang, Y., Zhang, X., Wang, C., Liu, J., & Ji, F. (2025). Understanding the interaction between generative AI and factual knowledge in supporting students' problem-solving. *Journal of Research on Technology in Education*, 1–23.
Rachha, A., & Seyam, M. (2024). LLM-enhancedlearning environments for cs: Exploring data structures and algorithms with gurukul. In *2024 IEEE frontiers in education conference (FIE)* (pp. 1–9). IEEE.
Rahman, M. M., & Watanobe, Y. (2023). ChatGPT for education and research: Opportunities, threats, and strategies. *Applied Sciences, 13*, 5783.
Ramirez Osorio, V., Zavaleta Bernuy, A., Simion, B., & Liut, M. (2025). Understanding the impact of using generative AI tools in a database course. *Proceedings of the 56th ACM Technical Symposium on Computer Science Education, 1*, 959–965.
Rane, N. L., Choudhary, S. P., & Rane, J. (2024). Gemini versus ChatGPT: Applications, performance, architecture, capabilities, and implementation. *Journal of Applied Artificial Intelligence, 5*, 69–93. https://doi.org/10.48185/jaai.v5i1.1052
Reihanian, I., Hou, Y., Chen, Y., & Zheng, Y. (2024). A review of generative AI in computer science education: Challenges and opportunities in accuracy, authenticity, and assessment. In *International conference on computational science and computational intelligence* (pp. 144–158). Springer.
Rincon-Flores, E. G., Castano, L., Guerrero Solis, S. L., Olmos Lopez, O., Rodríguez Hernández, C. F., Castillo Lara, L. A., & Aldape Valdés, L. P. (2024). Improving the learning-teaching process through adaptive learning strategy. *Smart Learning Environments, 11*, 27.
Roychowdhury, S. (2024). Journey of hallucination-minimized generative AI solutions for financial decision makers. In *Proceedings of the 17th ACM international conference on web search and data mining* (pp. 1180–1181).
Ruan, S., & Lu, K. (2025). Adaptive deep reinforcement learning for personalized learning pathways: A multimodal data-driven approach with real-time feedback optimization. *Computers and Education: Artificial Intelligence*, Article 100463.
Salas-Pilco, S. Z., Xiao, K., & Oshima, J. (2022). Artificial intelligence and new technologies in inclusive education for minority students: A systematic review. *Sustainability, 14*, Article 13572.
Samuels, A. J. (2018). Exploring culturally responsive pedagogy: Teachers' perspectives on fostering equitable and inclusive classrooms. *SRATE Journal, 27*, 22–30.
Sandhu, R., Channi, H. K., Ghai, D., Cheema, G. S., & Kaur, M. (2024). An introduction to generative AI tools for education 2030. *Integrating generative AI in education to achieve sustainable development goals*.
Sands, P. (2019). Addressing cognitive load in the computer science classroom. *ACM Inroads, 10*, 44–51.
Sarsa, S., Denny, P., Hellas, A., & Leinonen, J. (2022). Automatic generation of programming exercises and code explanations using large language models. *Proceedings of the 2022 ACM Conference on International Computing Education Research-, 1*, 27–43.
Schefer-Wenzl, S., Vogl, C., Peiris, S., & Miladinovic, I. (2024). Exploring the adoption of generative AI tools in computer science education: A student survey. In *Proceedings of the 2024 the 16th international conference on education technology and computers (ICETC '24)* (pp. 173–178). Association for Computing Machinery. https://doi.org/10.1145/3702163.3702188.
Schelhowe, H. (2004). Paradigms of computing science: The necessity for methodological diversity. *Gender, Technology and Development, 8*, 321–334.
Shin, J., Cruz-Castro, L., Yang, Z., Castelblanco, G., Aggarwal, A., Leite, W. L., & Carroll, B. F. (2024). Understanding optimal interactions between students and a chatbot during a programming task. In *2024 winter simulation conference (WSC)* (pp. 3106–3117). IEEE.
Shoaib, M., Hasnain, G., Sayed, N., Ghadi, Y. Y., Alajmi, M., & Qahmash, A. (2025). Automated generation of multiple-choice questions for computer science education using conditional generative adversarial networks. *IEEE Access*.
Sinha, A., Goyal, S., Sy, Z., Kuperus, R., Dickey, E., & Bejarano, A. (2024). Boilertai: A platform for enhancing instruction using generative AI in educational forums. *arXiv preprint arXiv:2409.13196*. https://arxiv.org/abs/2409.13196.
Styve, A., Virkki, O. T., & Naeem, U. (2024). Developing critical thinking practices interwoven with generative AI usage in an introductory programming course. In *2024 IEEE global engineering education conference (EDUCON)* (pp. 1–8). https://doi.org/10.1109/EDUCON60312.2024.10578746. Kos Island, Greece: IEEE.
Suárez-Pizzarello, M., SáNchez-Trujillo, M. D. L. A., & Flores, E. A. R. (2024). Exploring ChatGPT-4 as an academic assistant in thesis development: A case study on postgraduate higher education. In *2024 IEEE 4th international conference on advanced learning technologies on education & research (ICALTER)* (pp. 1–4). IEEE.
Sultana, K., & Sianaki, O. A. (2025). A multi-agent generative ai system for incorporating bloom's taxonomy, constructivism, and metacognition theories in student's learning performance. In *ACIS 2025 proceedings* (p. 134). Association for Information Systems. URL: https://aisel.aisnet.org/acis2025/134.
Susanti, W., Jama, J., Krismadinata, D. R., & Nasution, T. (2021). An overviuw of the teaching and learning process basic programming in algorithm and programming courses. *Turkish Journal of Computer and Mathematics Education (TURCOMAT), 12*, 2934–2944.
Sweller, J. (2020). Cognitive load theory and educational technology. *Educational Technology Research & Development, 68*, 1–16.
Takerngsaksiri, W., Warusavitarne, C., Yaacoub, C., Hou, M. H. K., & Tantithamthavorn, C. (2024). *Students' perspectives on AI code completion: Benefits and challenges. In 2024 IEEE 48th annual computers, software, and applications conference (COMPSAC)*. Osaka, Japan: IEEE. https://doi.org/10.1109/COMPSAC61105.2024.00252
Tankelevitch, L., Kewenig, V., Simkute, A., Scott, A. E., Sarkar, A., Sellen, A., & Rintel, S. (2024). The metacognitive demands and opportunities of generative AI. In *Proceedings of the 2024 CHI conference on human factors in computing systems* (pp. 1–24).
Tarun, B., Du, H., Kannan, D., & Gehringer, E. F. (2025). Human-in-the-loop systems for adaptive learning using generative ai. In *2025 IEEE frontiers in education conference (FIE)* (pp. 1–7). IEEE.
Tayeb, A., Alahmadi, M., Tajik, E., & Haiduc, S. (2024). Investigating developers' preferences for learning and issue resolution resources in the ChatGPT era. In *2024 IEEE international conference on software maintenance and evolution (ICSME)* (pp. 413–425). Flagstaff, AZ, USA: IEEE. https://doi.org/10.1109/ICSME58944.2024.00045.
Taylor, D. L., Yeung, M., & Bashet, A. (2021). Personalized and adaptive learning. *Innovative learning environments in STEM higher education: Opportunities, challenges, and looking forward*.
Tian, Y., Yan, W., Yang, Q., Chen, Q., Wang, W., Luo, Z., & Ma, L. (2024). Codehalu: Code hallucinations in LLMs driven by execution-based verification. arXiv preprint arXiv: 2405.00253 https://arxiv.org/abs/2405.00253.
Troussas, C., Papakostas, C., Krouska, A., Mylonas, P., & Sgouropoulou, C. (2024). Evaluating ChatGPT-driven automated test generation for personalized programming education. In *2024 2nd international conference on foundation and large language models (FLLM)* (pp. 194–200). IEEE.
Van Velzen, J. (2016). *Metacognitive learning: Advancing learning by developing general knowledge of the learning process*. Cham: Springer.
Wang, J., & Duan, Z. (2025). *Controlling large language model hallucination based on agent AI with langgraph. Preprint, Cambridge open engage*. https://doi.org/10.33774/coe-2025-xkwl5version1,postedJanuary13,2025.Contentnotpeer-reviewed. License: All Rights Reserved.
Weber, J. L., Neda, B. M., Juarez, K. C., Wong-Ma, J., Gago-Masague, S., & Ziv, H. (2024). Beyond the hype: Perceptions and realities of using large language models in computer science education at an r1 university. In *2024 IEEE global engineering education conference (EDUCON)* (pp. 1–8). IEEE.
Wei, S., Luo, Y., Chen, S., Huang, T., & Xiang, Y. (2023). Deep research and analysis of ChatGPT based on multiple testing experiments. In *2023 international conference on cyber-enabled distributed computing and knowledge discovery (CyberC)* (pp. 123–131). Jiangsu, China: IEEE. https://doi.org/10.1109/CyberC58899.2023.00030.
Winarta, I. M. W. P., Alfarozi, S. A. I., & Hidayah, I. (2024). Atomic evaluation using large language model for automated essay exam scoring. In *2024 IEEE international conference on communication, networks and satellite (COMNETSAT)* (pp. 21–27). IEEE.
Wu, Y. (2023). Integrating generative AI in education: How ChatGPT brings challenges for future learning and teaching. *Journal of Advanced Research in Education, 2*, 6–10.
Xu, X., Qiao, L., Cheng, N., Liu, H., & Zhao, W. (2025). Enhancing self-regulated learning and learning experience in generative AI environments: The critical role of metacognitive support. *British Journal of Educational Technology*.
Xue, Y., Chen, H., Bai, G. R., Tairas, R., & Huang, Y. (2024). Does ChatGPT help with introductory programming? An experiment of students using ChatGPT in cs1. In *Proceedings of the 46th international conference on software engineering: Software engineering education and training* (pp. 331–341).
Yan, W., Nakajima, T., & Sawada, R. (2024). Benefits and challenges of collaboration between students and conversational generative artificial intelligence in programming learning: An empirical case study. *Education Sciences, 14*, 433.
Yang, A. C., Lin, J.-Y., Lin, C.-Y., & Ogata, H. (2024). Enhancing python learning with pytutor: Efficacy of a ChatGPT-Based intelligent tutoring system in programming education. *Computers and Education: Artificial Intelligence, 7*, Article 100309.
Yatani, K., Sramek, Z., & Yang, C.-L. (2024). AI as extraherics: Fostering higher-order thinking skills in human-AI interaction. *arXiv preprint arXiv:2409.09218*. https://arxiv.org/abs/2409.09218.
Yeh, T. Y., Tran, K., Gao, G., Yu, T., Fong, W. O., & Chen, T.-Y. (2025). Bridging novice programmers and LLMs with interactivity. *Proceedings of the 56th ACM Technical Symposium on Computer Science Education, 1*, 1295–1301. https://doi.org/10.1145/3641554.3701867. New York, NY, USA: ACM.
Yilmaz, R., & Yilmaz, F. G. K. (2023). The effect of generative artificial intelligence (AI)-Based tool use on students' computational thinking skills, programming self-efficacy and motivation. *Computers and Education: Artificial Intelligence, 4*, Article 100147.
Yogarajan, V., Dobbie, G., & Keegan, T. T. (2025). Debiasing large language models: Research opportunities. *Journal of the Royal Society of New Zealand, 55*, 372–395.
Yu, Z., Liu, S., Denny, P., Bergen, A., & Liut, M. (2025). Integrating small language models with retrieval-augmented generation in computing education: Key takeaways, setup, and practical insights. *Proceedings of the 56th ACM Technical Symposium on Computer Science Education, 1*, 1302–1308. ACM.
Yusuf, A., Pervin, N., Román-González, M., & Noor, N. M. (2024). Generative AI in education and research: A systematic mapping review. *The Review of Education, 12*, Article e3489.
Zajda, J. (2021). Constructivist learning theory and creating effective learning environments. In *Globalisation and education reforms: Creating effective learning environments* (pp. 35–50). Springer.

Zhai, C., Wibowo, S., & Li, L. D. (2024). The effects of over-reliance on AI dialogue systems on students' cognitive abilities: A systematic review. *Smart Learning Environments, 11*, 28.
Zhang, Z., Dong, Z., Shi, Y., Price, T., Matsuda, N., & Xu, D. (2024). Students' perceptions and preferences of generative artificial intelligence feedback for programming. *Proceedings of the AAAI Conference on Artificial Intelligence, 38*, 23250–23258.
Zhang, Y., Li, Y., Cui, L., Cai, D., Liu, L., Fu, T., Huang, X., Zhao, E., Zhang, Y., Chen, Y., et al. (2023). Siren's song in the AI ocean: A survey on hallucination in large language models. arXiv preprint arXiv:2309.01219 https://arxiv.org/abs/2309.01219.
Zhou, X., Teng, D., & Al-Samarraie, H. (2024). The mediating role of generative AI self-regulation on students' critical thinking and problem-solving. *Education Sciences*.
Zimmermann, M., Janetzko, H., & Haymond, B. (2024). Integrating generative AI methods in computer science education: Perspectives, strategies, and outcomes. In *EDULEARN24 proceedings* (pp. 10358–10365). IATED.